

**DCTROY: OPTIMIZED FREQUENCY-DOMAIN IMAGE PROTECTION
USING WEIGHTED DCT BANDS WITH
WARM-START COLD-RESET ALGORITHM**

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DEDICATION

To God,

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CHAPTER I

INTRODUCTION

This chapter provides an overview of the research problem and objectives that are supported by the importance of the study, scope, limitations, significance, and definition of each term used in the study.

Background of the Study

The misuse of artificial intelligence-based image editing tools is no longer theoretical—it is already causing measurable harm. In a multi-country survey of over 16,000 respondents, 2.2% reported being directly targeted by non-consensual deepfake pornography, while exposure is broader: 15% of UK adults reported encountering harmful deepfakes involving impersonation, fraud, or explicit imagery (Umbach et al., 2024; Sippy et al., 2024). This victimization is enabled by widespread access to the technology. Numerous deepfake-capable models are publicly available, many specifically optimized for intimate imagery, making acquisition trivial for malicious actors (Hawkins et al., 2025). Yet human defenses are inadequate. In controlled experiments, participants correctly identified realistic deepfakes only one-third of the time (Lewis et al., 2023). This convergence—accessible tools, documented harm, and unreliable detection—exposes a fundamental asymmetry: exploitation outpaces defense. Diffusion-based editing tools, though neutral by design, are being systematically

weaponized for identity theft, non-consensual imagery, and fraud, motivating growing interest in defenses that operate before manipulation occurs.

This shift toward proactive protection is closely tied to the technical properties of modern diffusion-based generative models. Latent Diffusion Models (LDMs), such as Stable Diffusion, iteratively denoise compressed latent representations, enabling fine-grained manipulation through natural-language prompts (Rombach et al., 2022; Brooks et al., 2023). Personalization techniques, such as DreamBooth, allow identity-consistent generation from a small set of reference images (Ruiz et al., 2022). However, the intended ease of use brings risks of exploitation, including deepfakes, nude imagery generation, and other malicious manipulations.

To mitigate unauthorized or malicious image manipulation, several defense mechanisms have been proposed. Pixel-space defenses (e.g., PhotoGuard, AdvDM, Mist) directly alter RGB values but often involve trade-offs between effectiveness and perceptual quality. Additionally, latent-space defenses (e.g., Glaze, EditShield) perturb internal representations to improve stealth but remain vulnerable to purification and simple transformations (Devulapally et al., 2025). On the other hand, frequency-domain strategies (e.g., DCT-Shield, HF-ADB) aim to balance imperceptibility and robustness through structured spectral perturbations (Bala et al., 2025; Onikubo & Matsui, 2024). Despite the differences in their approach, most defenses rely on standard gradient-based optimization,

which is sensitive to initialization and often susceptible to getting stuck in the local minima due to the complex, high-dimensional nature of the loss landscape.

Unfortunately, persistent technical gaps remain. Existing methods provide inconsistent guidance on which frequency bands are most effective: HF-ADB emphasizes high-frequency components for style masking, whereas DCT-Shield reduces high-frequency energy to maintain imperceptibility and improve robustness against JPEG Compression (Bala et al., 2025; Onikubo & Matsui, 2024). Consequently, this lack of consensus indicates that rigid single-band strategies may be suboptimal, motivating adaptive or weighted frequency approaches. Additionally, optimization is further hindered by naive initialization, where starting with random noise often places the optimizer far from the effective adversarial manifold—the regions in perturbation space where small modifications induce semantically disruptive edits—resulting in slow convergence and high computational cost (Liu et al., 2020). Therefore, fixed-step gradient descent can become trapped, limiting the discovery of robust perturbations and producing highly variable outcomes across runs (Liu et al., 2020).

Consequently, to address the identified gaps, this study proposes a DCT-based adversarial perturbation framework that advances beyond existing approaches by considering both where to perturb and how to optimize them. Specifically, the proposed framework introduces a weighted frequency-band distribution of perturbation energy across all DCT components, addressing the

conflicting fixed-band strategies and ensuring an optimal balance of protection and imperceptibility. Additionally, a Warm-Start Cold-Reset algorithm is integrated to overcome the computational stagnation caused by naive initialization. Warm-start upscaling leverages coarse-to-fine optimization using gradients from lower-resolution perturbations to guide higher-resolution searches efficiently, while periodic cold-reset scheduling reinitializes optimizer momentum to mitigate stagnation and promote exploration beyond local minima. As a result, the proposed framework, encapsulating frequency-band weighting, warm-start upscaling, and cold-reset scheduling, aims to provide a more principled, technically rigorous approach for generating imperceptible yet effective protection against diffusion-based image editing.

Statement of the Problem

This paper seeks to address the following research problems:

1. What is the optimal perturbation distribution across different frequencies to achieve a balance between robustness, noise imperceptibility and edit protection effectiveness using frequency band weighting?
2. To what extent does the proposed framework improve upon established baselines in terms of edit protection effectiveness and imperceptibility, while maintaining its robustness under common image transformations and purification methods?

3. How does the proposed warm-start, cold-reset algorithm affect the convergence efficiency of the DCT perturbation optimization, considering the time and space computational cost?

Objectives of the Study

To ensure a systematic resolution of the technical limitations identified in the previous section, the objectives of this study are formulated to directly address the corresponding statement of the problem.

General Objective

The main objective of the study is to develop and validate an image adversarial perturbation framework that generates imperceptible perturbations in the frequency domain—incorporating color-channel prioritization and frequency-band weighting—using a warm-start, cold-reset algorithm, in order to disrupt semantic edits from diffusion-based editing models while maintaining robustness.

Specific Objectives

1. To determine the optimal perturbation distribution across different frequencies to achieve a balance between robustness, noise imperceptibility, and edit protection effectiveness using frequency-band weighting.
2. To evaluate the proposed method through controlled experiments that measure its effectiveness and compare it to baseline methods in terms of:

- a. Edit-protection quality (LPIPS, CLIPScore): measures the difference between edited protected and edited original outputs.
 - b. Perturbation imperceptibility (SSIM, PSNR, VIF): measures the visibility of the applied noise.
 - c. Robustness (LPIPS degradation): measure the resistance against JPEG compression, crop & resize, and purification.
- 3. To assess the proposed warm-start, cold-reset algorithm through controlled experiments that measure its efficiency and compare it to baseline methods in terms of:
 - a. Convergence performance (Relative Convergence Rate): measures how quickly the optimization achieves near-optimal protection
 - b. Runtime Overhead: measured as the average execution time per image
 - c. GPU Memory Overhead: measured as the peak GPU memory usage recorded during execution

Scope and Limitations

This study focuses on the design, development, and evaluation of a DCT-based adversarial protection method applied strictly at inference time, without poisoning training data or models. Furthermore, the experiments are conducted on 512×512 images under controlled conditions, using 500 images each from the publicly available InpaintCOCO and OmniEdit datasets, enabling reproducibility and standardized comparison. Additionally, the evaluation

emphasizes edit-protection effectiveness, perceptual imperceptibility, robustness, and convergence efficiency, using quantitative computational metrics such as: LPIPS, CLIPScore, SSIM, PSNR, VIFp, LPIPS Degradation, Relative Convergence Rate, average execution time per image and peak GPU memory usage recorded during execution, without relying on human perceptual evaluation.

The threat model is gray-box, reflecting realistic partial access to model information, and the study focuses on latent diffusion models subjected to instruction-based semantic editing and mask-guided inpainting attacks. Moreover, robustness is assessed under practical digital transformations, while complex recapture scenarios are outside the scope of the current experiments. In addition, evaluations are performed individually at inference without deployment pipelines or batch processing, and results are derived from controlled datasets rather than primary data collection. While the experiments concentrate on a specific generative architecture and attack setting, the methodological contributions—particularly the structured frequency-domain design and optimization strategy—are formulated as a general framework that may be adapted to other generative AI editing models in future work.

Significance of the Study

The outcomes of this study are expected to benefit multiple stakeholders in both societal and technological contexts:

End-users and content creators. The development of future tools that embed subtle, robust image protection through perturbations before upload might be of

service to individuals who share personal images on social media or creative platforms. This might reduce the risk of non-consensual generative edits, deepfakes, or automated scraping by requiring no significant changes in user behavior. Though these methods may not offer the same degree of effectiveness across different platforms and image processing pipelines, they can contribute to the improvement of user privacy and digital identity protection.

Digital platforms and technology providers. In the future, social networks, hosting services, and generative AI system developers may look into integrating frequency-guided image protection modules at upload or inference time. This will, in turn, aid in the mitigation of misuse, reduction of demands on moderation, and facilitation of compliance with new safety-by-design and content authenticity frameworks. The findings presented in this study may inform such integration by providing empirical insights into how identity-aware perturbations affect the outputs of generative models.

Government agencies and regulators. Public institutions developing standards for AI safety and digital content authenticity may gain evidence from this research in support of policy decisions on watermarking, image authenticity labeling, and privacy protection mechanisms. This may contribute to the potential development of national or regional guidelines addressing non-consensual image manipulation and generative model accountability in the future.

Enterprises and risk management teams. The findings could eventually be applied by organizations handling user imagery—be it identity verification,

e-commerce, or media platforms—to better safeguards against model-driven impersonation or manipulated content. It could help in the design of lightweight, privacy-preserving defenses that operate within practical runtime and resource constraints.

Researchers and the academic community. The study is expected to provide a reproducible framework that strategically determines where and how to perturb by combining color-channel prioritization and frequency-band weighting with an improved optimization pipeline—featuring warm-start upscaling and periodic cold-reset scheduling. This framework may be used as the foundation for further research in cross-model transferability, adaptive attacker response, and human perception analysis within the context of adversarial defenses for generative models.

Broader societal impact. The study might give a potential long-term contribution to understanding how adversarial perturbations balance the three elements of security, imperceptibility, and efficiency in a real-world setting. Such work could support efforts toward building public trust in and the integrity of media in the age of generative AI, aligning with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 16 (Peace, Justice, and Strong Institutions).

Definition of Terms

The following terms are defined conceptually and operationally as they are used in this study.

Adversarial Defense. Adversarial defenses are techniques designed to prevent or reduce unauthorized manipulation of machine learning models by introducing protective transformations or noise (Carlini et al., 2020). In this study, it refers to the DCT-based perturbation method applied before diffusion-model editing to shield facial identity from generative manipulation.

Adversarial Perturbation. Adversarial perturbations are small, carefully crafted modifications that alter a model’s prediction while remaining nearly invisible to humans (Croce & Hein, 2020). In this study, they are implemented as frequency-domain perturbations injected into quantized DCT coefficients to interfere with the VAE’s identity encoding.

Attack Success Rate (ASR). ASR is a metric that measures how often an adversarial attack successfully disrupts a target model (Carlini et al., 2020). In this research, ASR represents the proportion of perturbed images that successfully block identity-preserving edits during img2img and inpainting.

Coarse-to-Fine Optimization. Coarse-to-fine optimization is a hierarchical strategy that first adjusts low-frequency components and gradually refines high-frequency details (Fan et al., 2021). The study uses this approach to update low-frequency DCT bands before fine-tuning high-frequency perturbations.

Cold Reset / Reset Scheduling. Cold resets are optimization techniques that periodically reinitialize variables to escape local minima and stalled plateaus (Li et al., 2020). This study applies reset scheduling to prevent stagnation during perturbation optimization and improve convergence quality.

DCT Coefficients. DCT coefficients are numerical values produced by the Discrete Cosine Transform that encode image information across spatial frequencies (Aiazzi et al., 2021). In this research, perturbations are applied directly to 8×8 quantized JPEG-style DCT blocks to disrupt identity-relevant frequencies

Diffusion Models. Diffusion models generate images by reversing a forward noising process to reconstruct coherent visual outputs (Ho et al., 2020). This study targets the Stable Diffusion VAE encoder to disrupt how facial identity features are reconstructed during the reverse diffusion process.

Frequency Band. A frequency band is a grouped range of spatial frequencies, where low frequencies encode coarse structures and high frequencies encode fine textures (Rao & Hwang, 2020). In this study, different DCT frequency bands are manipulated selectively to balance imperceptibility and edit-protection effectiveness.

Gradients. Gradients are derivatives of the loss function used to determine the direction of parameter updates (Goodfellow et al., 2020). In this study, gradients guide the update of perturbations through backpropagation from the diffusion model.

High-Dimensional Curse. The high-dimensional curse describes difficulties in optimization when the number of parameters becomes excessively large (Shalev-Shwartz & Ben-David, 2020). It motivates the study’s use of warm-start initialization and reset scheduling to stabilize optimization in the high-dimensional DCT space.

Image-to-Image Inpainting. Inpainting refers to reconstructing or filling missing image regions using generative priors (Zhou et al., 2020). This research uses inpainting to test whether adversarial perturbations prevent identity-preserving reconstructions of facial areas.

Imperceptibility. Imperceptibility refers to the degree to which adversarial modifications remain invisible to human observers (Wang et al., 2021). In this study, it is evaluated through PSNR, SSIM, and LPIPS to ensure that perturbations do not visibly degrade the image.

Latent Diffusion Models (LDMs). Latent diffusion models operate on compressed latent representations instead of raw images to reduce computational cost (Rombach et al., 2022). The study uses LDMs, particularly Stable Diffusion, to test the robustness of the proposed perturbations.

Latent Space. Latent space is a compact encoding that preserves semantic structure and enables efficient manipulation (Chen et al., 2020). Here, it refers to the Stable Diffusion VAE’s latent z -space where identity features are targeted for disruption.

Local Minima. Local minima are points where optimization becomes stuck due to zero gradients, even when better solutions exist (Goodfellow et al., 2020).

This study employs cold-reset strategies to escape such regions during adversarial optimization.

LPIPS (Learned Perceptual Image Patch Similarity). LPIPS is a deep-learning-based perceptual similarity metric aligned with human visual judgment (Zhang et al., 2021). It is used here to measure whether perturbed images remain visually similar to their originals

Loss Landscape. A loss landscape is the geometric structure of a model’s loss function, containing flat, sharp, and unstable regions that affect optimization (Li et al., 2020). This study considers the loss landscape to design warm-start and frequency-aware update strategies.

Model-Agnostic. A model-agnostic method works across different architectures without requiring retraining (Tramèr et al., 2020). The proposed framework is model-agnostic because its perturbations are applied at the input level and generalize across diffusion models.

Projected Gradient Descent (PGD). PGD is an iterative constrained optimization algorithm that updates perturbations within an ϵ -bounded region (Rauber et al., 2020). It serves as a baseline against which the proposed perturbation method is compared.

Transformation Robustness. Transformation robustness is the ability of adversarial perturbations to remain effective under image transformations like

resizing or compression (Wu et al., 2020). This study tests robustness through JPEG recompression, scaling, and purification.

VAE Encoder / Decoder. The VAE encoder compresses images into latent codes while the decoder reconstructs them (Kingma & Welling, 2021 summary). In this study, perturbations are injected during encoding and evaluated after decoding.

Warm Start. Warm starts to initialize optimization using previously effective parameter states to accelerate convergence (Li et al., 2020). This study uses warm starts to stabilize perturbation generation and reduce computation time.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the theoretical framework, and relevant literature and studies that the researcher considered in strengthening the importance of the present study. It also presents the conceptual framework to fully understand the research for better comprehension of the study.

Theoretical Framework

This section explores the foundational theories, models, algorithms, and key concepts that ground the study. Figure 1 illustrates the diagram of the theoretical framework of the study.

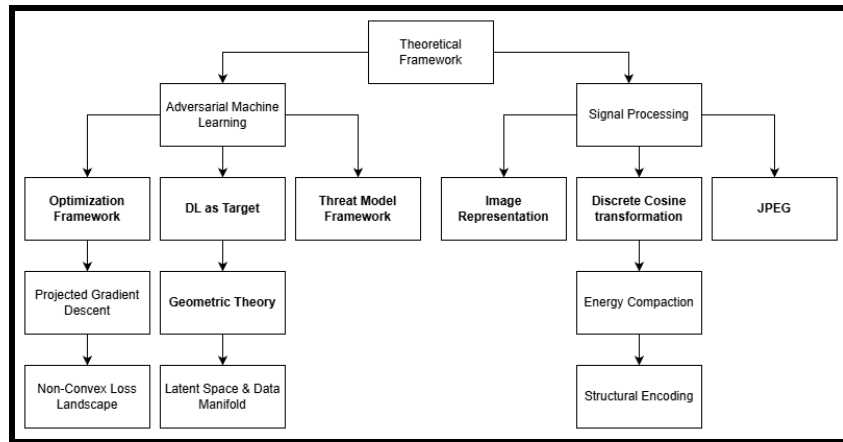


Figure 1

Theoretical Framework of the Study

I. Adversarial Machine Learning

Adversarial Machine Learning (AML) is formally defined by the National Institute of Standards and Technology (NIST) as the study of attacker capabilities

and the design of algorithms to resist security challenges (NIST AI 100-2, 2023). Unlike standard machine learning, which focuses on average-case generalization, AML addresses the worst-case behavior of algorithms in the presence of malicious inputs.

Optimization Framework

A key theoretical idea in adversarial machine learning is the Min–Max Optimization Framework from Madry et al. (2018). This framework views adversarial robustness not as an accidental weakness, but as a formal optimization problem with two competing goals. It is expressed mathematically as:

$$\min_{\theta} \rho(\theta) \quad \text{where} \quad \rho(\theta) = E_{(x,y) \sim D} \left[\max_{\delta \in S} L(f_{\theta}(x + \delta), y) \right]$$

- The **inner maximization** represents the adversary: it tries to find a perturbation (within an allowed set S) that maximizes the model’s loss.
- The **outer minimization** represents the defender: it tries to find model parameters that minimize the worst-case loss caused by such perturbations.

Deep neural networks make this problem difficult because their loss surfaces are non-convex. That means the loss function is a high-dimensional, bumpy surface with many local maxima, saddle points, and irregular curves. In such a landscape, simple maximization methods can get stuck on smaller local peaks and fail to find the true “worst-case” perturbation described in the inner maximization.

To handle this, Projected Gradient Descent (PGD) is utilized to approximate the inner maximization. It works by:

- Taking iterative gradient steps to climb the loss surface.
- After each step, projecting the perturbation back into the allowed set S .

Hence, PGD is theoretically regarded as the “universal first-order adversary”—a standard method to test whether the Min–Max objectives are effectively satisfied in a complex, non-convex loss landscape.

Manifold Assumption Geometric Framework

The Manifold Assumption, as investigated by Gilmer et al. (2018) and Stutz et al. (2019), provides a geometric explanation for the existence of adversarial examples. The central premise is as follows:

- High-dimensional natural images (points in $\mathbb{R}^d$) do not uniformly occupy the entire space. Rather, they are concentrated on a lower-dimensional, structured manifold.
- Neural networks, in contrast, often learn decision boundaries that extend substantially beyond this manifold into the broader, undefined “ambient space.”

Adversarial attacks exploit this structure by pushing data points off the natural manifold:

- Natural images occupy a compact, low-dimensional region.
- Perturbations orthogonal to the manifold lead to representations in the ambient space.
- In this space, the model’s learned boundaries become unstable or poorly defined.

The geometric idea of the manifold is realized in practice through the concept of latent space. In deep learning, the latent space is a compressed, lower-dimensional representation of data, where the structure of the manifold is explicitly captured

Theoretical research indicates that a model’s robust generalization relies on its ability to correctly map inputs onto this latent manifold. In this context, adversarial perturbations can be interpreted as attempts to distort this mapping, causing the model to extract incorrect latent features from the input.

Threat Model Framework

For the purpose of systematically assessing the adversarial robustness of machine learning models, the Adversarial Threat Model Taxonomy is used by the researchers, which was first suggested by Carlini and Wagner (2017) and further elaborated by NIST (2023). A threat model determines the adversary’s restrictions and the capabilities, hence, effectively marking the borders of the optimization problem that was described earlier (Akhtar et al., 2021).

The adversarial knowledge aspect stipulates the amount of information that the adversary possesses concerning the target model f . NIST differentiates between three levels:

- **White-Box:** Total model access (includes architecture, weights, and gradients); allows very accurate gradient-based attacks (the worst-case scenario).
- **Gray-Box:** Some partial knowledge (such as architecture and data distribution) but no access to the trained weights; usually a surrogate model is employed in order to approximate the gradients.
- **Black-Box:** No internal access whatsoever; only the model can be queried and the outputs observed; gradients have to be either estimated or transferred.

Conversely, the enemy is limited to making changes that will keep the input's semantic meaning intact. L_p norms are used to impose this condition, as they determine the distance between the original input x and the adversarial example x' . Usual perturbations are often restricted to an ϵ ball:

$$\|x' - x\|_p \leq \epsilon$$

- **L_∞ Norm (Maximum Norm):** Limits the maximum change allowed to any single feature (e.g., pixel), preventing noticeable alterations.

- **L_2 Norm (Euclidean Norm):** Limits the overall magnitude of the perturbation, allowing small changes across many features while avoiding large, concentrated modifications.

Furthermore, the adversary's objective defines the desired outcome of the attack:

- **Untargeted Attacks:** Success occurs if the input is classified as any class other than the correct label y , i.e., $f(x') \neq y$.
- **Targeted Attacks:** Success occurs only if the input is classified as a **specific target class** y_{target} , i.e., $f(x') = y_{target}$.

II. Digital Signal Processing

As mentioned in the Digital Image Processing (4th Edition), in the case of digital image processing, a natural image is represented mathematically as a two-dimensional discrete signal consisting of spatial coordinates that are variable. The modeling of images as signals allows for the strictest application of signal processing theory to be done in a systematic way through analysis, transformation, and manipulation of visual data.

Spatial-Domain Representation

The spatial domain visualizes an image directly via pixel intensities, with each pixel denoting a specific coordinate. In the case of color images, this expands to a multi-channel depiction of red, green, and blue intensities:

$$f[m, n] = [f_R[m, n] f_G[m, n] f_B[m, n]] \in R^{M \times N \times 3}$$

Spatial-domain processing works through local neighborhood transformations, such as filtering or convolution, which are applied at pixel level. Although it seems intuitive, perturbations in pixel space have the disadvantage of being brittle: small, curated changes can lead to misclassification while being visually indistinct at the same time.

Frequency-Domain Representation

Fourier theory establishes that any 2D signal can be decomposed into sinusoidal basis functions oscillating at distinct spatial frequencies. The 2D Discrete Fourier Transform (DFT) defines this mapping:

$$F[u, v] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m, n] e^{-j2\pi\left(\frac{um}{M} + \frac{vn}{N}\right)}$$

with inverse transform:

$$f[m, n] = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F[u, v] e^{j2\pi\left(\frac{um}{M} + \frac{vn}{N}\right)}, \quad j^2 = -1$$

Frequency interpretation for adversarial defense:

- **Low-frequency** components ($|u|, |v| \ll M/2, N/2$) encode global structure, smooth gradients, and macroscopic shapes; they are perceptually critical.

- **High-frequency** components encode edges, fine textures, and noise; they are perceptually expendable and can be strategically modified to disrupt models while preserving human perception.

Discrete Cosine Transform (DCT)

Conceding that the Discrete Fourier Transform (DFT) gives a theoretical basis, the Discrete Cosine Transform (DCT) acts as the practical foundation in image processing and defense techniques. In conformity with Rao and Yip (1990) and the JPEG standard, its real-valued coefficients correspond to common image formats and compression pipelines, besides the energy compaction that distributes the visual information mostly to the low-frequency components.

The 2D DCT of an $M \times N$ image block $f[m, n]$ is:

$$C[u, v] = \alpha_u \alpha_v \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m, n] \cos\left(\frac{\pi(2m+1)u}{2M}\right) \cos\left(\frac{\pi(2n+1)v}{2N}\right)$$

for $u \in \{0, \dots, M - 1\}, v \in \{0, \dots, N - 1\}$, with normalization constants:

$$\alpha_u = \left\{ \sqrt{\frac{1}{M}}, u = 0, \sqrt{\frac{2}{M}}, u > 0 \right\}, \quad \alpha_v = \left\{ \sqrt{\frac{1}{N}}, v = 0, \sqrt{\frac{2}{N}}, v > 0 \right\}$$

Inverse DCT reconstructs the spatial image:

$$f[m, n] = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} \alpha_u \alpha_v C[u, v] \cos\left(\frac{\pi(2m+1)u}{2M}\right) \cos\left(\frac{\pi(2n+1)v}{2N}\right)$$

The Discrete Cosine Transform (DCT) offers many significant features that render it the best choice for frequency-domain adversarial defense. To begin with, it demonstrates the property of **energy compaction** by directing the major part of an image's visual content to low-frequency coefficients while the high-frequency components mostly consist of edges, textures, and noise. Besides, this divide results in a natural axis for perturbations aimed at a specific point, which permits the alterations in high-frequency areas to be concealed visually and at the same time the structure of the whole image remains unaltered.

Trailing on this, **quantization** makes use of the uneven distribution of energy as a basis for its operation; thus it is able to perform a very rough approximation of high-frequency coefficients and to frequently round them off to zero. The incorporation of such a controlled loss constitutes a marker for the robustness of the system against perturbation. Due to this, only the most effective adversarial changes that withstand quantization will be able to use the compression that is lossy and still impact the image like JPEG.

Inherent in the DCT process is **frequency decomposition**, which results in the global structure being totally separated from the fine details. The global and the noisy parts of an image cannot be separated through pixel-domain operations, but in the transform domain, it is possible to control the frequency bands independently. Along with the energy compaction and quantization, this allows for very accurate structural control which permits the creation of noise that is not

perceived but is enough to disrupt the representations of the model, thus creating the theoretical basis for the frequency-domain defense strategies.

JPEG Compression

The images that are uploaded to online platforms are generally subject to automatic compression, which is done mainly to lessen the requirements of storage and bandwidth. The JPEG (Joint Photographic Experts Group) lossy image compression standard for photographs is the most commonly used one, utilizing block-based processing and frequency-domain transformation in order to cut down the size of files while maintaining their quality in terms of the human perception (Wallace, 1992). A block of 8×8 pixels is processed through DCT, quantization, and entropy encoding to achieve a reduction in file size that is efficient and at the same time perceptually quality-wise the same.

One of the characteristics of JPEG is it removes or reduces high-frequency components, which are often the ones that hold the subtle details. Consequently, images uploaded on the internet have been through imperfect yet close reconstructions where minor scale variations have been lost, which is a trait that becomes very crucial in case of image perturbations, watermarking, or even adversarial defenses. JPEG, being the standard, has the ability to handle low and mid-frequency components more and this has made it possible for the frequency-domain operations that utilize these bands to keep the visual fidelity even after the lossy compression has taken place. This characteristic not only sets

a realistic operational limit but also provides a basis for designing perturbations that are durable against the typical online image transformations.

A related advancement transforming traditional JPEG for machine learning contexts is the differentiable approximation introduced by Reich et al. (2024), which overcomes the non-differentiable barriers of standard JPEG—such as zero gradients from rounding, flooring, and clipping in quantization—to enable seamless integration into deep learning pipelines. Unlike conventional JPEG, where discrete operations block backpropagation, this method uses polynomial rounding ($x \approx x - x^3$), shifted approximations for floor/scaling, and soft clipping, achieving superior fidelity and effective gradients validated via stronger adversarial attacks. This justifies its relevance for frequency-domain studies like adversarial defenses, as it allows precise simulation of compression effects during optimization, ensuring perturbations remain robust across real-world quality variations.

Literature Review

The following literature review provides a comprehensive analysis of the current state of adversarial defense mechanisms for image protection.

Diffusion Models, Stable Diffusion, and Identity-Driven Editing

Generative models for image synthesis have gained popularity because of their capacity to generate content that is not only high-quality but also very realistic. The major developments in the field like BigGAN, StyleGAN, and diffusion models have gradually resulted in higher image fidelity and resolution. LDMs, in particular, project images into a latent space of lower dimension concentrating on the most pertinent semantic features thereby lessening the computational burden (Rombach et al., 2022; Bonechi et al., 2025).

Stable Diffusion (SD), which is a popular Latent Diffusion Model(LDM), utilizes a U-Net backbone together with a latent VAE encoder for the generation of images based on textual prompts. The newer Stable Diffusion XL (SDXL) not only boosts this feature to producing images of higher resolution (up to 1024×1024 pixels) but also includes conditioning on crop sizes and aspect ratios, making it possible to exercise finer semantic control (Podell et al., 2024).

In the real world, SD has already been used for various purposes like personalization according to the subject and image-to-image editing. For instance, DreamBooth (Ruiz et al., 2023), which manages to edit a model that preserves identity, with the help of a textual token, by fine-tuning a diffusion model with just 5-10 reference images. Likewise, InstructPix2Pix (Brooks et al., 2023) provides natural-language-based editing without the need for per-image fine-tuning. The methods discussed are indications that SD is capable of producing images that keep the main identity traits intact through the editing process. On the other hand,

GANs are very much limited by their purity because they have to deal with the trade-offs between getting good reconstructions and editability. While the use of a modular adapter for fine-tuning in SD is commonplace, GANs suffer from a very serious phenomenon known as catastrophic forgetting, wherein the training of the model on a particular subject affects its overall knowledge negatively. As a result, in order to maintain the identity of the subject of the highest quality, it normally requires costly, per-image optimization (like Pivotal Tuning Inversion) instead of a single, efficient fine-tuning pass (Ali et al., 2025; Robb et al., 2020).

SD's latent representations have semantic information. Bonechi et al. (2025) conducted a study on the features acquired from the U-Net of the SD and discovered that the lower-resolution layers (16×16) are responsible for capturing the structural and semantic features. Even under simple image transformations, such as rotation, flipping, scaling, cropping, and adding a little noise, the resulting representations remain relatively stable. The fact that they are stable implies that SD's latent space is able to pass through the identity-relevant information, which is important for both generative editing and the understanding of privacy issues.

Privacy Considerations, Membership Leakage, and Motivation for Defense

As much as SD's latent representations allow for flexible editing and anonymization, they can still leak identity-related data. A recent study by Duan et al. (2023) proved that it is possible to conduct membership inference attacks on

LDMs, including SD v1.4 and v1.5, at a high level of success. Their SecMI technique reveals that some latent-space features contain enough identity information to determine if a dataset was used for fine-tuning, even in large-scale pretrained models. Therefore, these results highlight that one of the drawbacks of editing with latent-space representations is privacy issues, and these issues should be addressed.

Besides, researches implied that SD can be used in situations related to identity manipulation. Among these was the work of Mehta et al. (2024), where it was proposed to use LDFaceNet (a latent diffusion-based framework for high-fidelity face swapping) as an example of how diffusion models can make identity-preserving edits in tightly controlled experimental conditions that look very realistic. In a similar vein, Sun et al. (2024) also proved that Stable Diffusion could be applied in the area of deepfake detection by means of feature reconstruction, thus, emphasizing the model's significance for both generation and forensic contexts. Both cases point out that SD's latent-space potentials are not limited to deepfakes or identity-centered manipulation, but also to other areas where such methods are not necessarily abused.

Simultaneously, SD has been employed in privacy-preserving situations. The RAD framework (Malm et al., 2024) adopted SD and ControlNet to create anonymized visuals that maintain the integrity of the context as well as the

effectiveness of the downstream tasks, proving that the opacity of the representations can be utilized for protection as well.

Inclusively, these results bring the twofold aspect of Stable Diffusion: its latent-space representations harness and empower the generative and anonymization functions. But at the same time, they provide the means for the creation of deepfakes, impersonation, and theft of identity, thereby necessitating the invention of strong protections that would cover latent-space weaknesses.

Latent-space features carry both semantic and identity information, but most defense approaches work right from very basic pixel level operations, where the images are simply changed to make it more difficult for the attackers to guess whether they are part of the training dataset or the model's output. Pixel-space defenses are particularly relevant because diffusion models ultimately generate images from pixel data, and subtle manipulations in this space can disrupt reconstruction attempts or obfuscate identity information. Nevertheless, Xue and Chen (2024) noted that the process of denoising could reduce the effects of the perturbations applied on the pixel level. This indicates that relying solely on these defenses might not be enough in the case of attacks aimed at latent representations. Hence, the comprehension of pixel-space defenses is the first and necessary step before proceeding to the more sophisticated latent-space protections.

Pixel-Space / RGB-Based Defenses

The activation of threats raised by diffusion models has led to the consideration of various defense mechanisms among researchers. The use of passive methods like watermarking or the addition of provenance metadata results in a facilitation of traceability while the protection against identity manipulation and forgery is not much through these methods. On the other hand, active defenses intervene directly by introducing perturbations (noise) that are meant to disrupt the model's ability to reconstruct or mimic identity features. This section outlines the strengths and weaknesses of the existing pixel-based defenses.

The earliest active defense, **PhotoGuard** (Salman et al., 2023), implements the baseline pixel perturbation strategy that disturbs the unauthorized manipulation of images with the help of diffusion models, formulated as:

$$\delta_{diffusion} = \arg \min_{\|\delta\|_{\infty} \leq \epsilon} \|f(x + \delta) - x_{targ}\|_2^2.$$

It works by the addition of imperceptible noise directly and the application of L_{∞} constraint. The technique's adversarial perturbations injected into RGB values have a budget of $\epsilon=16/255$, and it applied the perturbations over 200 optimization steps. The quantitative evaluation of the method in terms of FID scores ranks from 130.6 to 167.6, while for SSIM it is between 0.50 and 0.58 and for PSNR up to 14.91 and down to 13.58 indicating that the protected and unprotected images are vastly different. The "noise" produced acts as constrained adversarial examples that are aimed for diffusion models' latent space encoder and they are taking VAE encoder, used in latent diffusion pipelines, as their weak point.

The main shortcoming of PhotoGuard is its sensitivity to the common image transformation. The authors of the paper revealed that the transformations involved in JPEG compression, rescaling, and noise addition can easily render the protection ineffective or at least lessen it. Hence, this issue limits the use of the method in real life where the transformations are always taking place, especially on the internet. The method used would lead to a potent disruption if used in controlled conditions, but its effectiveness would drop instantly if the standard post-processing operations were operated on the images. Thus, this defense has limited its use in privacy protection that is required in real life. The work establishes a fundamental approach, but at the same time, it emphasizes the necessity for resilient methods that withstand real-world image transformations.

AdvDM (Liang et al., 2023) takes the basic idea and enhances it through iterative optimization by adding noise in the form of adversarial images defined as $x' = x + \delta$, where δ is limited to the pixel space. To perform the defense, AdvDM uses “unbounded” noise of the image-grade gradient descent method to the loss of the diffusion models, which is already in pixels, and thereby tricks the whole diffusion process. This hardware creates copies of the original images via the expression $x' = x + \delta$, where modifications δ are connected to the pixel space and limited by the ℓ_∞ norm. Alongside this, the iterative optimization takes place in pixel-space where the gradients are computed in relation to the pixels at every step, thereby making the pixel-space the clear area of perturbation. AdvDM has

made it clear through experimentation that the FID score for LSUN-sheep has been bumped up from 32.81 to 203.5 when hiding the pictures in a manner that their quality drops from 0.64 to 0.01. A similar increase in FID is seen for LSUN-cat and LSUN-airplane with the score going up to 127.04 and 169.67 respectively. The optimization is carried out over 5,000 iterations along with 40 Monte Carlo sampling steps to fix the perturbation budgets at $8/255$ in the ℓ_∞ norm.

In spite of prevailing disruptive effects in controlled evaluations, the research indicates that defense power suffers severely with JPEG compression: FID is reduced from 169.67 to 61.67 after compression, which indicates a big loss of security. Furthermore, the technique does not assure any robustness against real-world image pipelines where compression and resizing are performed as standard preprocessing steps. AdvDM offers a powerful pixel-level defense in a controlled, white-box setting but still points to its weaknesses when it comes to robustness against preprocessing and strategic allocation of perturbation effort.

Extending this study, **MIST** (Liang & Wu, 2023) is directed to provide pixel-based adversarial examples that keep the security of images throughout different pipelines. The approach blends two pixel-space loss functions: a semantic loss (interrupting the diffusion model's usual training process) and a textual loss (maximizing encoder distance to a given target image). All adversarial perturbations are kept within limits and applied straight in the pixel domain

performance boosts provided by fused loss are reported for Van Gogh artworks during textual inversion: FID rises from 237.56 (no attack) to 465.82 with fused weight $w=104$, and precision decreases from 0.90 to a minimum of 0.02. Under DreamBooth, the fused mode secures FID of 444.92 (in contrast to the 274.40 baseline) while precision is brought down to 0.10 (from 0.88). The technique proves to be more transferable than the approaches using single losses and thus shows effectiveness in all in (L_∞ -bounded, using projected gradient descent). Ththe mentioned situations: textual inversion, DreamBooth, and image-to-image.

Effectiveness is noted to be highly sensitive to hyperparameters—particularly the weight given to loss terms and the choice of target image. For instance, images with high contrast ratio and repetitive patterns (like the "MIST" logo pattern) produce stronger attacks and more robustness against transformations, while images with low contrast yield weaker results. Simple input transformations such as crop-and-resize and noise addition can weaken protection effectiveness. The authors caution that "the performance of Mist does not strictly follow a consistent pattern with changes in the fused weight," indicating that interactions between the two loss terms may not be fully predictable. They explicitly advise that "larger-quantity experiments are needed for a deeper understanding." MIST extends pixel-space defenses with improved transferability across multiple DM-based imitation pipelines, but remains challenged by parameter selection sensitivity and varying robustness to

transformations. The requirement for careful tuning of the fusion weight and target image selection limits its practical applicability in deployments where such customization is infeasible.

Progressing towards efficiency and visual fidelity, **Diff-Protect** (Xue et al., 2024) delves into the underlying design bottlenecks of pixel attacks and latent diffusion models. The study disclosed that the adversarial perturbations in the latent (z) space might be around 10 times greater than those in the pixel (x) space, which indicates that the encoder is much more prone to attack than the denoising module. Instead of using the full-gradient PGD method, the researchers switch to Score Distillation Sampling (SDS), which is an approximation that eliminates the gradient term $\partial z_t / \partial \epsilon$ and consequently lowers the computational costs intensely. This reduces the amount of memory used from ~16 GB to ~8 GB and the duration of processing from ~65 seconds to ~30 seconds, thus achieving a 50% reduction in both areas while still keeping the effectiveness of the protection. For SDEdit protection, SDS(-) (gradient descent on semantic loss) performs 0.698 in SSIM, 29.562 in PSNR, and a competitive FID between 206.01 and 231.57 for different edit strengths. Human evaluations indicate that SDS(-) outputs are judged as more natural (HumanEval score of 4.55) compared to MIST (2.44) and AdvDM (2.94), which means that the protection perturbations are hardly perceptually noticeable.

However, this approach is only evaluated on LDMs only, while pixel-space architectures are excluded. Furthermore, although SDS is a technique that offers

benefits in terms of computation, still the method is working within the pixel-space perturbation paradigm and does not eliminate the basic weaknesses to transformations or purification that have been the case with the earlier pixel-based methods. Through focusing the optimization on the latent encoder and extinguishing computational inefficiency through SDS, Diff-Protect has made a significant contribution to the latent diffusion models, which is, to realizing the simultaneous gain in speed and perceptual quality improvement. Nonetheless, the limitation of the model to LDMs and absence of any evaluation on other generator architectures are the factors that restrict the generalizability.

DiffusionGuard (Choi et al., 2025) revolutionizes the pixel-space paradigm by letting adversarial perturbations only occur in regions defined by the user, formulated as:

$$\delta = \arg \max_{\|\delta\|_{\infty} \leq \eta} L_{adv}(\theta; x + \delta, M_{tr}) = E_{M \sim A(M_{tr})} \left[\|\epsilon_{\theta}(x_T; y_{text}, T, M, x + \delta)\|_2^2 \right].$$

By doing so, it specifies the early stage of the diffusion process where global structure is being established. The procedure pushes the norm of the predicted noise to the maximum at the first step of the denoising process, thereby selecting the early moments when inpainting models create fine details. Besides, it exploits mask augmentation to come up with strong perturbations that can be effective through different mask shapes. Each of the perturbations is allowed to exist in the specified mask region only, and their magnitude is controlled by ℓ_{∞} limits (for

instance, 16/255). In the performance evaluation on InpaintGuardBench (42 pictures, 5 masks each, 2,100 total edit tasks), DiffusionGuard scores SSIM values of up to 0.990 and PSNR values of up to 41.58, thus proving that it not only retains good-quality images but also offers protection. When it comes to the masks that were already seen, DiffusionGuard offers a PSNR of 12.60, while the unseen masks give 13.19; thus the latter performance demonstrates even more robustness. The assessments conducted with human participants indicate that DiffusionGuard has a 41.9% probability of winning over PhotoGuard with 25% of the outcomes being equal on the unseen masks.

Tests are conducted primarily on Stable Diffusion Inpainting v1.4 and demonstrate some transferability to Stable Diffusion 2.0 Inpainting. However, the evaluation scope is limited to open-source, inpainting-focused diffusion models. Additionally, while mask-augmentation improves robustness, the method's effectiveness depends on the assumption that attacker and defender share understanding of the sensitive region. DiffusionGuard demonstrates effective, mask-localized pixel perturbation defense for inpainting scenarios, showing empirical protection and perceptual subtlety under tested conditions. However, its applicability is constrained by the inpainting-specific evaluation and unproven generalization to broader editing paradigms.

Additionally, **AntiPure** (Yang et al., 2025) proposes pixel-space adversarial perturbation designed to survive powerful "purification" (diffusion-based denoising) that precedes malicious customization in generative workflows, addressing the two-stage purification-customization (P-C) pipeline. The method embeds L_∞ -bounded noise directly in the input image and employs two guidance mechanisms: Patch-wise Frequency Guidance (PFG), using Discrete Cosine Transform (DCT) to locally strengthen high-frequency components in image patches, and Erroneous Timestep Guidance (ETG), which distributes adversarial impact across multiple diffusion timesteps by minimizing prediction discrepancies. Jointly, Projected Gradient Descent is used for optimization, and both the guidance terms are first weighted and then exponentially scaled. From a quantitative standpoint, AntiPure demonstrates remarkable post-purification resilience: on VGGFace2 after DreamBooth fine-tuning with the application of 20 purification iterations, AntiPure delivers FID=59.97, ISM (Identity Score Matching)=0.6271, and BRISQUE=44.63, which are the results that no other methods like Anti-DB or SimAC can surpass. Furthermore, AntiPure results in only slight perceptual-change scores: VGG-LPIPS=0.3884, showing that perturbations are still undetectable even though their protection is highly effective.

Efficacy is heavily accustomed on the PGD budget, the selection of erroneous timestep t_{err} , and the guidance hyperparameters λ_1 and λ_2 . Also, the method demands white-box knowledge of the purification details. There is no

testing of robustness against more intricate or adaptive purification algorithms. AntiPure demonstrates that defenses in the pixel space, when directed by frequency and temporal cues, can withstand the cleansing process, thus achieving both invisibility and distortion. Nevertheless, its diagnostic range, need for white-box conditions, and dependence on detailed environmental knowledge restrict its use as a general defense that is practically applicable.

All of the previously mentioned pixel-based defense methods have several common limitations that occur consistently. Their defense is often diminished by standard image transformations, many rely on non-selective pixel-space perturbations that do not take into account semantic or frequency structure, and most are extremely dependent on the choice of hyperparameters. On top of that, each technique has limited usage—be it restricted to certain types of models, certain editing pipelines, or presumptions of white-box access. The overall picture painted by these limitations is that current pixel-domain defenses are still not robust and universal enough for image use in the real world under uncontrolled conditions. Moreover, this combined drawback indicates the requirement of new defense tactics, perhaps in other representation domains or through the design of more adaptive systems.

Latent/Diffusion-Aware Defenses

The researchers subsequently looked into the latent domain which was represented by the methods such as EditShield and Glaze. However, at the same

time, another research in diffusion-based adversarial purification, exemplified by DiffPure, unwrapped the crucial fragility in latent-domain defenses. In this approach, the internal representations of the model are changed rather than the pixels, resulting to very high degree of imperceptibility. Nonetheless, they still come under heavy attack from frequency-domain preprocessing and diffusion-based purification, and are accompanied by a major requirement of computational and architectural dependencies that are very restrictive for real-world deployment.

As mentioned by **EditShield** (Chen et. al., 2023), where they changes the course of the instruction-based image editing by moving VAE latent encodings through a regulated $4/255$ perturbation budget, making the changes unnoticeable by using L2-norm constraints, concurrently offering robust protection through various editing styles and instruction phrases. The optimization process involves finding solutions through iterative projected gradient descent for each image, and although very useful in laboratory settings, the security measures are strictly limited to certain model types (InstructPix2Pix v1.5 and MagicBrush-fine-tuned ones). Not only that, the method is indeed dependent on the architecture. It considers having white-box access to the particular VAE encoder and diffusion model being attacked and its effectiveness significantly decreases when editors use different architectures not specifically accounted for during the optimization process.

The main weakness is the vulnerability of the protective signal to frequency-domain purification. The adversary's use of low-pass filtering, Fourier-based compression, or cutting out high-frequency noise selectively at the input to the VAE encoder leads to a situation where the noise created is reduced in its effect without any noticeable change to the image. EOT models account for only spatial-domain transformations like Gaussian smoothing, rotation, cropping, and hence do not include frequency-selective preprocessing that keeps the image semantics intact while removing the protection signals.

Furthermore, **GLAZE** (Shan et. al., 2025) has presented a defense that is a combination of style transfer and feature-space optimization. This dual defense reported being able to gain 93.3% and more artist-rated protection success while also having an efficiency of more than 90.2%, even when different feature extractors are used. By contrast, the system is very expensive in terms of computation. For instance, the making of a single cloak involves the generation of style transfer and 500 iterations of optimization that take an average of 1.2 minutes on a high-end Titan RTX GPU and 7.3 minutes on a consumer CPU plus per artwork. The cost of computation coupled with dependency on particular model parts makes the implementation hard for artists having large stockpiles of paintings, especially those with hundreds or thousands of pieces that need protection.

In addition to cost, GLAZE is dependent on the architecture and necessitates white-box access to the feature extractor (Φ) that was used during the optimization. Nevertheless, the authors display their proof of robustness across various extractors (Φ -A, Φ -B, Φ -C) and thus claim a great generalization ($>90.2\%$ success), protection does get impaired a lot if the extractors are significantly different in terms of training data or architecture. The core issue is still left unsolved: the GLAZE’s pixel-domain LPIPS limits that are focusing on the visible spatial distortions and the feature-domain optimization which is depending on the latent shifts within the VAE are completely opposite to each other. Frequency-selective operations like low-pass filtering, high-frequency attenuation, and Fourier compression that operate orthogonally to both layers selectively weaken the perturbation signals before the VAE processing without triggering the pixel-domain detection. The wide-ranging assessments ensure robustness against the threats from the spatial domain (Gaussian noise $\sigma=0.15$, JPEG compression quality=10, pixel smoothing—each with protection degradation $<5\%$), but unfortunately, they do not provide any critical evidence regarding the frequency-domain preprocessing evaluation.

Nie and colleagues (2022) created **DiffPure** as an adversarial defense for classifiers, which led to them attaining the highest level of robustness in the industry (+5.44% on CIFAR-10 AutoAttack ℓ_∞ , +11.31% against BPDA+EOT) by making use of the forward and reverse SDE processes. However, they did not

realize that their invention also revealed a major problem for image protection techniques, which depending on perturbations in the latent-space. On the contrary, DiffPure's noise-based purification process, i.e., diffuser, regards every minor perturbation signal as noise, either adversarial or protective, and thus will systematically smoothen and recover-noise. This, in turn, reveals the inherent weakness of EditShield and GLAZE. If denoising is used, the visible and unprotected perturbations created in the latent space will be purified and, therefore, the defense will be made ineffective without the image showing any signs of being altered.

The technique entails the simultaneous resolution of two stochastic differential equations (one forward and one reverse) during each inference, leading to the inference time that is directly proportional to the diffusion timestep: around 5.1 seconds at $t^*=0.05$, 10.5 seconds at $t^*=0.1$, and 15.3 seconds at $t^*=0.15$ for each image processed on an NVIDIA V100 GPU. This increase in computation of 100-200 times over standard classification renders the deployment in real-time quite difficult. More importantly, the performance of DiffPure highly depends on the architect of the classifier (ResNet, WideResNet, ViT) and the method necessitates a pre-trained diffusion model that is specific to each dataset (e.g., Score SDE for CIFAR-10, Guided Diffusion for ImageNet, DDPM for CelebA-HQ). The defense is not transferable across different architectures without

retraining or adapting the model, indicating the same architectural brittleness that EditShield and Glaze suffer from.

The concealed flaw reflects both techniques for protecting images: DiffPure presumes that during the process of forward diffusion, adversarial examples still maintain their initial frequency composition. The use of frequency-selective preprocessing, such as low-pass filtering, might severely reduce or even completely not detectable by the human eye perturbation components whilst keeping the semantic part of the input intact. Thus, the adversarial signal is further reduced before $x(t^*)$ computation. So, this limits the noise structure the reverse SDE has to eliminate, thus, weathering the theoretical assumptions regarding noise removal. The current threat assessments (AutoAttack, BPDA+EOT, StAdv) are quite lacking in that they do not consider frequency-domain preprocessing, which indicates an untested area that is common to all three methods working in the latent-space.

Latent-space/diffusion-based techniques—EditShield, GLAZE intended for image protection, and DiffPure aimed at adversarial perturbation purification—display a common critical architectural ignorance. They all consider that protective or adversarial signals survive to the latent stage of processing with frequency-domain integrity still intact. Nevertheless, none thoroughly assesses the robustness against frequency-selective preprocessing.

These techniques are fragile for alternative models, require a lot of computations, and are still not tested against frequency-domain preprocessing that might weaken signals without leaving any traces. EditShield needs an iterative optimization for each image and presumes white-box access to particular VAE encoders and Glaze takes about 1-7 minutes per artwork and relies on certain feature extractors. At a more basic level, the very existence of DiffPure as a defense mechanism proves the point that diffusion-based purification has the potential to become a weapon against image protection techniques—any protective perturbation that is small enough to be unnoticeable can be treated as adversarial noise and thus be easily purified.

These limitations point out that the latent-space techniques move the vulnerabilities around instead of getting rid of them. They are able to remove the visibility in the pixel space, but at the same time, they introduce fragility in the latent space that affects both frequency-domain preprocessing and attacks purifying through diffusion. The overlapping of these weaknesses points to the requirement for the investigation of defenses working in the frequency domain in order to tackle threats that neither latent- nor pixel-based methods can completely overcome.

Frequency Defense

Despite these advances on latent and pixel-based defenses, they still struggling. This leads to open a new paradigm to use frequency in adversarial perturbation. These techniques by focusing on or incorporating protective signals in the frequency spectrum want to survive both spatial and latent-domain transformations, hence, providing a more robust protection against the ever more sophisticated image manipulations.

The **DCT-Shield** method (Bala et al., 2025) successfully tackles the frequency-domain weakness that was identified in EditShield, GLAZE, and DiffPure, by working right in the Discrete Cosine Transform (DCT) domain. Unlike pixel- or latent-space techniques whose signals can easily be deleted by frequency-aware operations, DCT-Shield take advantage of the JPEG pipeline, where it changes the quantized DCT coefficients with noise that survives compression-based purification. This method guarantees the detection of protective signals even when they are subjected to aggressive JPEG or other frequency-based preprocessing.

This approach attains excellent protection along with near-invisibility, boasting FID values of 35.023 for noise perception and LPIPS of 0.684 for edit protection, together with a human evaluation score of 4.22 which is better than the scores of the baselines including SDS(-), PhotoGuard, MIST, and AdvDM. It has been tested with different JPEG quality levels (65–95%) and has consistently shown its strength while at the same time, pixel-space defenses have proven to be

ineffective. Besides, it can easily switch between text-guided edits, inpainting areas, or adjusting the luminance (Y) channel, thus, regulating the tradeoff between imperceptibility and robustness, which cannot be achieved with previous pixel-domain techniques.

DCT-Shield, in contrast to a large number of latent-space defenses, is highly efficient in terms of its computational requirements. In addition, it is also agnostic to the type of model being used, and only involves the optimization of the VAE-encoder without an U-Net dependency. Moreover, it utilizes $\sim 50\%$ of the parameters that pixel-based methods do. It has been tested against various types of purification strategies, such as JPEG compression, cropping and resizing, and AdvClean filtering, and has been able to provide very good quality images with very slightly increased LPIPS scores (0.15–0.25 over competitors) even at the compression quality of ≤ 75 . These outcomes confirm that the frequency domain is a very powerful and, therefore, a dominant approach against preprocessing-aware attacks compared to the spatial or latent methods.

Although DCT-Shield JPEG-aware optimization has already achieved a robust defense against attacks in the spatial domain, **SimAC** (Wang et al., 2024) has gone a step further to apply the frequency-domain defense right through the diffusion process. It implies that the effectiveness of adversarial noise is a function of both the timing (timesteps) and the place (decoder layers) where perturbations are applied, which results in a very high amount of identity disruption through the

use of adaptive greedy timestep selection and feature-aware optimization. SimAC uses adaptive greedy search to discover frequency-relevant timesteps and then selectively disrupts structural and textural features at those times to prevent the creation of a unique identity for the attacker in customization attacks. The former methods would randomly add noise across timesteps, thereby, making the effort on high-timesteps where gradients are nearly zero, because the diffusion models process high-frequency (textural) information at low timesteps and low-frequency (structural) components at higher timesteps. Through targeting lower timesteps and deeper U-Net decoder layers, SimAC ensures that the adversarial perturbations are aimed at the features that are most significant for identity representation, thus enhancing the efficiency of both optimization and protection.

Empirical results validate considerable gains as compared to Anti-DreamBooth: The Face Detection Failure Rate of 9–19 percentage points (87–97% versus 64–77%) is a clear indicator of identity protection, while matching the Identity Score one's decrease by 0.16–0.20. These increments happen with new prompts and model versions (Stable Diffusion v1.4 and v2.1) as per the same model used. SimAC produces these gains just by encoder-based optimization, hence it does not need full diffusion backpropagation and thus is computationally efficient and suitable for diverse diffusion architectures. Moreover, qualitatively, SimAC+Anti-DreamBooth is found to induce more facial distortion than the baselines but the image quality is overall retained. The claim of

SimAC is that the temporal perception of the diffusion models is dynamic and the spatial perception is heterogeneous in different decoder layers. DCT-Shield is based on spatial frequency decomposition, whereas SimAC uses the gradient effectiveness and hierarchical feature representation, both being dependent on the specific timestep and perturbations being aligned with the model’s frequency-sensitive stages. Contrastingly, limitations include high dependence on particular diffusion architectures, the need for white-box access, attack being specific to identity-based customization, and per-image computational overhead due to adaptive greedy search. It also potentially opens up for manipulations outside its frequency-aware timestep selection, which is a point that shows how frequency-domain optimization increases robustness but cannot completely eliminate vulnerabilities.

HF-ADB (Onikubo & Matsui, 2024) changes the approach from SimAC’s temporal frequency optimization to spatial frequency decomposition and is based on the latter. The new approach is to separate the high-frequency and low-frequency bands and to apply significantly stronger perturbations to the high-frequency regions. This tactic not only reduces the impacts of style and facial mimicry, but it also does not make the noise easy to detect, thus able to withstand both the conventional filters and the diffusion-based purification attacks. HF-ADB is the answer to the weakness of the former defenses because the purification methods like DiffPure or bilateral filtering have been able to remove the

perturbations so far. By making a spatial decomposition of the images, HF-ADB applies strong perturbations to the high-frequency regions (which include edges and textures and typically account for 3–5% of the whole image) and weak perturbations to low-frequency regions, thus maintaining the non-detectability. The high-frequency elements are important for the identification and style-detection process, but they take up only a few spatial areas, thus making the targeted perturbation both cost-effective and potent. The technique employs a 3×3 Laplacian filter to pinpoint the high-frequency regions and then applies different levels of perturbation to the masked and unmasked ones, thus extending the ASPL from Anti-DreamBooth by this crucial masking step.

The empirical studies confirm that the HF-ADB technique is still effective on the part of the defense where the Anti-DreamBooth method does not succeed. When it comes to the purification of the bilateral filter, the Anti-DreamBooth’s FDSR is 0.97 with BRISQUE 42.3, but HF-ADB has FDSR 0.96 and BRISQUE 51.5 on the other hand. When compared with DiffPure, the Anti-DreamBooth method fails completely (BRISQUE 10.7), on the contrary, HF-ADB keeps BRISQUE 8.3, thus preserving the unnatural patterns and stopping the facial mimicry. The generation of the adversarial noise takes around 2 minutes per image that is running on an NVIDIA A100 GPU, which is about the same as the time taken by the baseline methods. The success of the HF-ADB is credited to its high-frequency robustness exploitation: these elements are very important for

identity/style, very difficult to remove without causing visual degradation and also very sparse hence the perturbation can be put in a concentrated manner. The purification techniques that are meant to remove low-frequency perturbations automatically leave the high-frequency noise intact, thus keeping the defense alive. However, there are still some limitations; it is possible that the robustness may be a concern in the case of imperceptibility, since the strong high-frequency noise might be noticeable when viewed up close, and the performance is scenario-specific—effective for images of faces but considerably weaker for artwork where low-frequency information might be the one dominating style encoding.

Conversely, recent research argues that there is no agreement on the best frequency bands for adversarial perturbations among low, mid, or high, as different results are reported depending on objective, dataset, and model architecture. Luo et al. (2022), emphasizing the invisibility of perturbations, impose a “low-frequency constraint” which forces perturbations into the high-frequency range, claiming that high-frequency noise is less easily perceived by human observers. On the other hand, Guo et al. (2020) prove that the success of the attack and the cost of the query are lower when using low-frequency perturbations, and this is so even if the image is subjected to common transformations that typically suppress high-frequency noise. This disagreement, as depicted in Figure 2 above, is rooted in the DCT where low-frequency coefficients (top-left) represent robust

global structure while high-frequency coefficients (bottom-right) correspond to perceptually expendable details and noise.

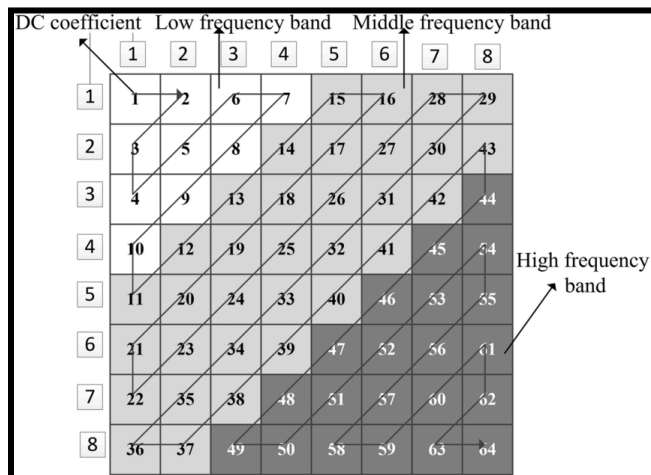


Figure 2

DCT Frequency Band Distribution

The issue is made even more complex by the fact that different researchers (Maiya et al., 2021) have declared there to be no universally perfect band at all, as they observed that the concentration of effective perturbations changes a lot from one dataset to another (e.g. CIFAR-10 vs. ImageNet). Chen et al. (2025) have just opened a new aspect of the debate by introducing architectural considerations, where they determined that Vision Transformers—an essential part of Latent Diffusion Models—are overly susceptible to low-to-mid frequencies which is a complete opposite to the high-frequency bias typical of CNNs. All these contradictory results notwithstanding, the current state of the art is still predominantly focused on classifier-based evasion attacks with hardly any

consideration given to the applicability of the conflicting principles to generative-model defenses. The present study’s proposal of a frequency-band weighting scheme that simultaneously utilizes the imperceptibility advantages of high frequencies and the robustness properties of low frequencies is thus driven by the need for synthesis.

Chances are that you are referring to an image's frequency representation which is the main culprit and offering the least noticeable noise in return for robust protection. Apart from the fact that it is a very difficult task to find the right places to perturb, the real trick is how to best optimize those perturbations as it is still very hard to deal with.

Dominant Optimization Strategy: PGD and Its Limits

Defenses working in the frequency domain are mainly dependent on the regular gradient-based methods like Projected Gradient Descent (PGD), but still, adversarial optimization is quite challenging mainly due to three structural limitations: the very high-dimensional search space, dependency on initialization, and complicated loss landscapes. The current section discusses the issues and the aforementioned limitations resulting in the current defense technologies, describes the shortcomings of conventional optimization and finally supports the need for an optimization workflow tailored to the frequency domain.

In Table 1, the defense mechanisms against image manipulation in diffusion-based models are summarized. A very noticeable tendency can be observed among these methods: almost all of them based on gradient-based optimization, especially Projected Gradient Descent (PGD) and its various forms. The core optimization for PhotoGuard, AdvDM, MIST, EditShield, GLAZE, DCT-Shield, SimAC, and HF-ADB is based on PGD. Even though Diff-Protect and DiffusionGuard are somewhat different, they still employ gradient descent mechanisms that are very similar to PGD principles—Diff-Protect makes use of SDS-style approximation, whereas DiffusionGuard utilizes early-stage perturbation targeting.

Table 1

Comparison Table of Existing Methods

Method	Domain	Optimizer Type
PhotoGuard	Pixel	PGD: (Encoder + Diffusion)
AdvDM	Pixel	PGD + Monte Carlo
MIST	Pixel	PGD Fused Loss
Diff-Protect	Pixel	PGD Semantic Latent
DiffusionGuard	Pixel	PGD Early-Stage + Mask Aug
EditShield	Latent	PGD VAE Latent (L2+LPIPS)

Table 1*Comparison Table of Existing Methods (cont'd)*

Method	Domain	Optimizer Type
GLAZE	Latent	PGD Style Transfer
SimAC	Pixel (Frequency-Guided)	Adaptive PGD + Features
HF-ADB	Pixel (Frequency-Guided)	PGD High-freq Bands
DCT-Shield	Frequency	PGD DCT Coefficients

Even though the methods differ greatly, they still have three main limitations in common. The fact that they are widely used reveals both the ease of current optimization methods and their main drawback: all these methods depend on the same type of fixed-step, single-scale, or domain-agnostic optimization that cannot handle the high-dimensional non-convex perturbation landscapes. This calls for the development of more robust, scalable, and model-agnostic optimization methods that can help defenses effectively.

PGD Bottleneck

The common reliance on hyperparameters that cannot be changed, the optimization at a single scale, and the non-adaptive PGD-style updates are the main sources of the restrictions inherent in all existing methods. The mentioned

optimizers have tough times in the multiparameter spaces of perturbation where gradients are noisy, and local minima are plenty. The later iterations often produce no improvement at all and yet are costlier to compute. Furthermore, the attacks are often initiated by random or naïve means, which results in them being far from the regions of the loss landscape containing the most information, thus, prolonging their convergence and increasing the chances of their being caught in flat or misleading regions—so it is clear that the success and efficiency of the strong perturbation finding process are greatly affected by the starting point choice.

Naive Initialization Constrains Convergence in Adversarial Perturbation Generation

High-dimensional perturbation spaces are a cause of a significant limitation not only due to their size, but also because the loss landscape becomes very complicated. Hence, it prevents gradient-based approaches from roaming freely among the different minima.

The initialization method has a direct impact on how quickly and efficiently the adversarial perturbation can be generated. PGD-Imp (Li et al., 2024) proves that the dynamic step size allocation—beginning with a small size and then gradually increasing it—takes an average of 34.2 iterations to achieve high attack success as against 100+ for the conventional PGD on ResNet-50, and this all with

a runtime cut from 428s to 57s. LoRa-PGD (Savostianova et al., 2024) supports that warm-start initialization with a prior PGD step using truncated SVD does not robust accuracy (stronger attack effectiveness) under equal 10-step budgets, indicating that warm-start placement produces better perturbations than random starts. RWR-NM-PGD (Li et al., 2021) shows that the optimization is stuck in local optima when the initialization is set to zero. The authors claim that using random restarts along with the I-FGSM method increases the attack success rate by 27.19% while doing so with the standard PGD increases it by 9.27%.

The overall conclusion to be drawn from these findings is that naive initialization, whether it be zero or random kind, does not allow convergence and hence keeps the optimization from reaching regions of better solutions. In contrast, warm-start and curriculum-based strategies optimize closer to the productive areas of the loss landscape, which leads to either faster convergence or higher-quality perturbations under equivalent computational budgets. In this connection, Topal et al.(2024) show that attacking a downscaled version of a high-resolution image and then “blowing up” the resulting noise is, in effect, a warm-start for high-dimensional perturbation spaces: the low-resolution attack very efficiently finds the right gradient directions and the upscaled perturbation gives a good initialization for the high resolution space. This strategy further reinforces the concept that the proper placing of the initial perturbations can avoid some of the problems posed by high-dimensional landscapes. Still, even with these enhanced

initialization techniques, the optimization in high-dimensional adversarial spaces is still prone to local minima.

Local Minima Susceptibility in High-Dimensional Adversarial Optimization

In the study of Liu et al. (2020), they scrutinize and delineate the adversarial training landscape. The first major finding is that the Hessian eigenvalues go higher and higher as the adversarial budgets increase, which means sharper minima are present. The second one is that gradients do not completely disappear during the training—they are still quite large instead of being small, which implies that the trajectory bounces rather than settles. Lastly, in the case of adversarial training, the minima are isolated. This indicates that if you have two models trained independently, then there is no smooth, low-loss path connecting their solutions—any route between them will have to go through areas of higher loss. This phenomenon is also more pronounced in high-dimensional spaces, where the number of easy "escape routes" is limited, thus making it harder for optimizers to switch between good solutions.

The aforementioned restrictions are the direct reason for the slow convergence and high computational cost which have been shown in all the defenses in Table 1. However, in order to tackle the optimization difficulties, Liu et al. suggest Periodic Adversarial Scheduling (PAS), a method that alternates the adversarial strength during the training process. PAS enables the optimizer to

escape from getting stuck in sharp or disconnected minima. Along with that, it also enhances convergence from suboptimal initializations and it is less sensitive to hyperparameter choice, which together results in more stable and effective training than standard adversarial methods.

Synthesis

Generative models, especially Latent Diffusion Models (LDMs) like Stable Diffusion (SD), have made it possible to synthesize and edit images of very high quality and at the same time preserve the identity of the original (Rombach et al., 2022; Bonechi et al., 2025; Podell et al., 2024). Correspondingly, SD can be quickly and easily fine-tuned for subject-driven personalization using merely 5-10 reference images, which allows the making of identity-preserving edits using the help of textual tokens (Ruiz et al., 2023; Brooks et al., 2023). This relative ease of SD fine-tuning is one of the features turning it into a dual-use tool: it is beneficial for creative and privacy-preserving purposes (Malm et al., 2024), but at the same time, it might help the perpetration of unethical acts such as the generation of deepfakes or identity manipulation, all with very little data and computational resources (Duan et al., 2023; Mehta et al., 2024; Sun et al., 2024).

Moving on, the study of image safeguarding in generative models has developed over three main directions. To begin with, pixel-space defenses such as PhotoGuard, AdvDM, MIST, Diff-Protect, DiffusionGuard, and AntiPure

manipulate the RGB or image patches to prevent unauthorized reproduction or editing. Although these techniques make for good disruption in lab settings, their perturbation fades easily with the common transformations like JPEG compression, image resizing, and noise addition (Salman et al., 2023; Liang et al., 2023; Liang & Wu, 2023; Xue et al., 2024; Choi et al., 2025; Yang et al., 2025). Latent-space or diffusion-aware defenses, i.e., EditShield and GLAZE, which are the main representatives of this line of research, perturb the VAE or feature-space representations. The two approaches offer the same advantages of high imperceptibility and protection in white-box situations. However, they are more costly in terms of computations, architecture-dependent, and susceptible to purging methods like DiffPure whose the same method can remove latent perturbations while leaving the images unseen (Chen et al., 2023; Shan et al., 2025; Nie et al., 2022). At last, frequency-domain defenses like DCT-Shield, SimAC, and HF-ADB are the ones extending the protection either in the DCT frequency domain or adaptive timesteps. By dealing with DCT coefficients, high-frequency parts, as well as layer-specific attributes, these approaches generally reinforce their defense against both preprocessing and latent-space attacks, thus attaining less detectability and better protection than pixel- or latent-based methods. However, they still require a large amount of computation and are usually specific to a certain domain (Bala et al., 2025; Wang et al., 2024; Onikubo & Matsui, 2024).

This synthesis makes it clear that even with the progress in frequency-domain defenses, there are still some persistent issues that have not been tackled in the literature. These challenges not only reveal the drawbacks of the existing methods but also encourage more research to be done:

1. **Frequency Band Conflict.** A major challenge remains disagreement about the frequency bands that provide the most protection. High-frequency changes normally lead to better undetectability (Luo et al., 2022) while low-frequency changes increase the success and durability of the attack (Guo et al., 2020). Besides, it differs from one architecture to another. For example, Vision Transformers in Latent Diffusion Models are more sensitive to low-to-mid frequencies than CNNs (Chen et al., 2025). Nevertheless, there is still no such thing as a universal frequency-band weighting framework that incorporates both effectiveness, robustness and imperceptibility.
2. **Naive Initialization.** Ultimately, the setup of perturbations is another limitation. The practice of random or zero initialization drives the optimization to uninformative areas. This prolongs convergence and diminishes the effectiveness of the attack (Li et al., 2021; Li et al., 2024; Savostianova et al., 2024). Warm-start and curriculum-based techniques have been demonstrated to speed up convergence and bolster attack potency in high-dimensional spaces (Topal et al., 2024).

3. **Local Minima Susceptibility.** A number of defenses rely on either the standard PGD or its variants (Salman et al., 2023; Liang et al., 2023; Chen et al., 2023), which have a tendency to converge very slowly and at times get stuck at local minima that are not optimal (Liu et al., 2020). This points out the requirement for optimization strategies that are adaptive and can move through complex search spaces in a better way.

Finally, the thorough understanding of the related research and the identification of their major technical weaknesses set the basic premise of this study. The recognized gaps give the main empirical support for a new intervention. Therefore, this research becomes the theoretical basis for the new study, which is directly applied to the development of a defense mechanism. This new mechanism not only relies on established principles but also tackles the ongoing shortcomings of the current technology.

Table 2

Contribution Scope of DCTroy

Scope	PhotoGuard	Diffusion Guard	MIST	DCTroy
Frequency Domain	✗	✗	✗	✓
Frequency-Band Weighting	✗	✗	✗	✓
Instruction-based	✓	✗	✓	✓

Table 2*Contribution Scope of DCTroy (cont'd)*

Scope	PhotoGuard	Diffusion Guard	MIST	DCTroy
Inpainting	✗	✓	✗	✓
Standard PGD	✓	✓	✓	✗
Random Initialization	✓	✓	✓	✗

Literature Map

A literature map is displayed in Figure 3 to create an image of the literature review landscape and pinpoint the exact location of this research. The studies discussed are associated with diffusion models, threat models, and the current works are classified in the figure into defenses based on Pixel-Space, Latent-Space, and Frequency-Domain.

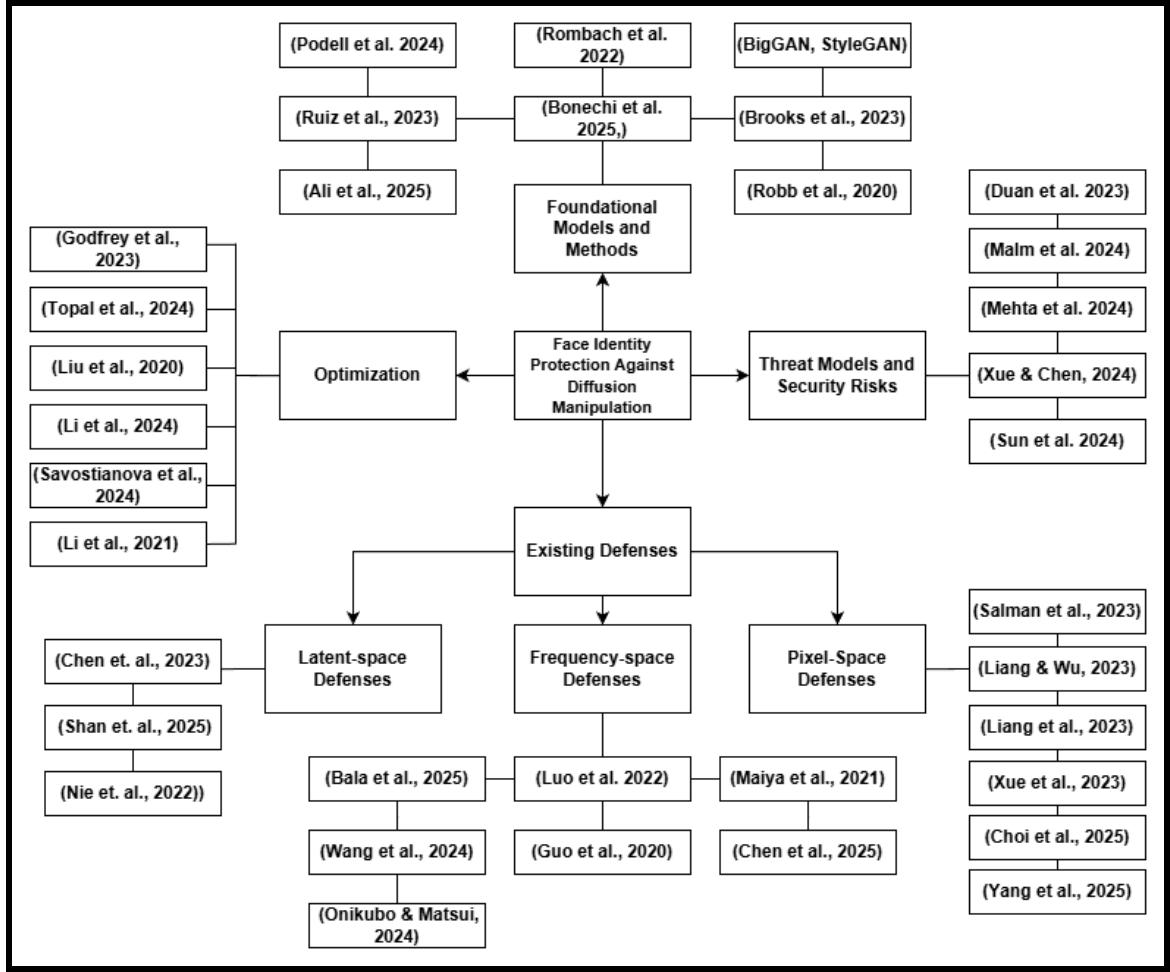


Figure 3

Literature Map of Related Literature

Conceptual Framework

In order to critically view the whole process of turning the identified technical gaps into validated contributions. The diagram illustrated in Figure 4 sets out the logical path of the research and shows how established theoretical foundations are blended to overcome certain optimization bottlenecks. The framework, by clarifying the links between the problem space, the theoretical

kernel and the proposed solution, offers a detailed plan for the development of the artifact and its later thorough evaluation.

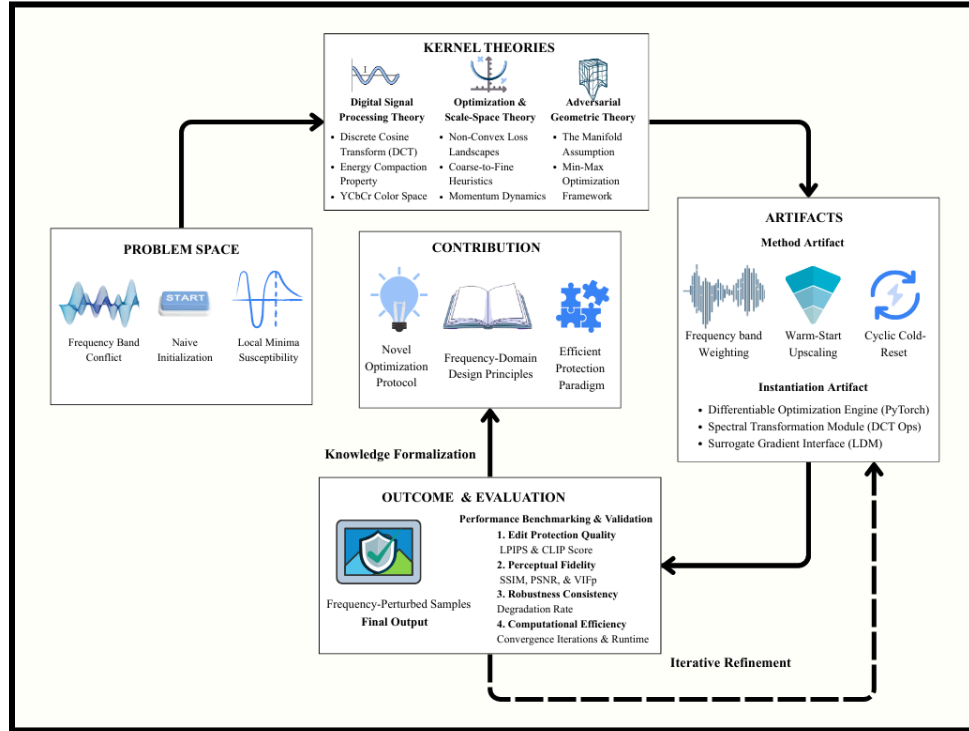


Figure 4

Conceptual Framework of the Study

This diagram outlines that the whole process starts from the Problem Space, where the gaps on frequency band conflict, naive initialization, and susceptibility to local minima continue to exist. Kernel theories involve applying Digital Signal Processing (DSP), non-convex optimization dynamics, and manifold theory to create the Method Artifact and its respective Instantiation Artifact. Later on, Performance Benchmarking measures the performance of the developed solution based on four main criteria, namely, edit protection, imperceptibility, robustness,

and efficiency. These take place within an Iterative Design Refinement feedback loop, where validated results would align with the objectives of the study. Subsequently, the final Knowledge Contributions are established, in particular, the discovery of a novel frequency-aware optimization protocol that allows for effective and robust adversarial defense.

CHAPTER III

RESEARCH DESIGN AND METHODOLOGY

This chapter presents the research design, methodology, and technical processes planned to be followed during the course of the study. Specifically, the entire structure and justification of the suggested framework, the experimental setup and the evaluation strategies will be described. Furthermore, the study will specify the tools, benchmark datasets, and baselines that will be used during the research process.

Research Design

This research utilizes the Design Science Research (DSR) methodology (illustrated in Figure 5)—a methodical procedure for creating as well as evaluating computational artifacts which solve real technical problems. Design Science Research is suited for this study because it offers strict guidelines for the creation of an innovative algorithm, system, or framework, showing its usability through experimental implementation, methodically assessing its power against predetermined baselines, and progressively refining both the artifact and the knowledge of the base. The DSRM process consists of steps that are sequential and yet iterative, giving the researchers various points of entry and loops for iterations. This allows them to go back to prior steps in light of evaluation results or new ideas. Hence, it not only supports the creation of the artifact but also the

knowledge generation during the design, demonstration, evaluation, and refinement cycle.

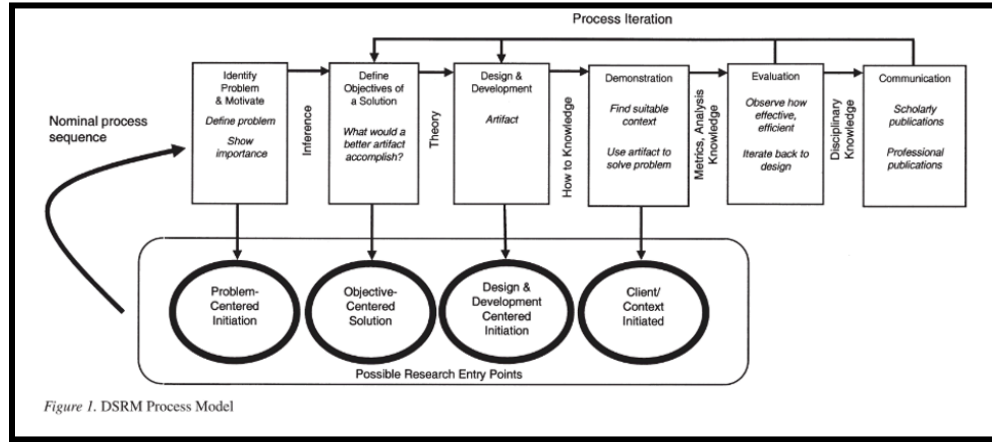


Figure 5

Design Science Research Methodology (Peffers et al., 2007)

Design and Development of the Artifact

The design and development stage of the study explains the artifact, which is the proposed DCT-based adversarial perturbation framework, to be conceptualized, organized, and operationalized. Therefore, this section presents the method utilized, the basic design principle, and the formal definition of the algorithm.

Problem Formulation

The purpose of the method is to create a protected image that can withstand identity-aware editing assaults, and at the same time, it should not be detectable

when compared to the original picture. Formally, let an original RGB image be represented as $x \in \mathbb{R}^{H \times W \times 3}$; then the objective is to determine a distortion of the DCT domain that, once applied and decoded back to the pixel space, results in an image x' such that:

- Edits applied to x' visually differs from edits applied to x (edit protection effectiveness)
- x' is perceptually similar to x (noise imperceptibility)
- This protection is preserved under common transformations (robustness).

Inspired by the frequency-domain formulation of DCT-Shield (Bala et al., 2025), the proposed method also explicitly integrates the JPEG pipeline into the optimization. Given an input image x and JPEG quality setting Q_{alg} , the patch-wise **quantized DCT coefficients** is computed first using a JPEG encode operator $JPEG_E$:

$$\alpha = JPEG_E(x; Q_{alg}), \alpha(c) \in \mathbb{Z}_p^{n_p(c) \times 8 \times 8},$$

where $c \in \{Y, Cb, Cr\}$ indexes the luminance and chrominance channels, and $n_p(c)$ denotes the number of 8×8 blocks in channel c . The operator $JPEG_E(\cdot)$ represents the full sequence from RGB–YCbCr conversion to block-wise DCT and quantization.

Let α_Y , α_{Cb} , and α_{Cr} denote the quantized coefficients for the Y, Cb, and Cr channels, respectively. The proposed method perturbs only the luminance channel, introducing an additive perturbation δ_Y in the quantized DCT domain:

$$\tilde{\alpha}_Y = \alpha_Y + \delta_Y.$$

The protected image $x'(\delta_Y)$ is then obtained by combining the perturbed luminance coefficients with the original chrominance coefficients and applying the JPEG decode operator $JPEG_D$:

$$x'(\delta_Y) = JPEG_D(\tilde{\alpha}_Y, \alpha_{Cb}, \alpha_{Cr}; Q_{alg}).$$

Modern diffusion-based editors (e.g., Stable Diffusion) operate in the latent space of a pre-trained VAE. Let $E(\cdot)$ denote this VAE encoder. The proposed method attacks the encoder by optimizing a norm minimization loss on the latent representation of the protected image:

$$L(\delta_Y) = \|E(x'(\delta_Y))\|_2^2$$

Intuitively, this loss encourages the VAE to encode $x'(\delta_Y)$ into an atypical latent representation that leads to distorted or inconsistent edits, while still allowing $x'(\delta_Y)$ to remain visually close to x in pixel space.

The base optimization problem is therefore:

$$\delta_Y^* = \underset{\delta_Y}{\operatorname{argmin}} L(\delta Y)$$

subject to the following constraints.

L_∞ magnitude and JPEG-quantization constraints

To keep perturbations small and visually unobtrusive, δ_Y is constrained inside an L_∞ ball of radius ϵ :

$$\|\delta_Y\|_\infty \leq \epsilon,$$

where ϵ controls the maximum allowed change to any Y-channel DCT coefficient. Because α_Y consists of quantized coefficients, perturbations smaller than one quantization level may be completely removed by subsequent JPEG compression.

The proposed method uses a frequency weighting over the Y-channel DCT coefficients. A fixed weight tensor W_{freq} assigns a scalar weight to each DCT frequency within an 8×8 block, explicitly redistributing the allowable perturbation magnitude across frequency bands by amplifying or suppressing perturbations according to the predefined weighting scheme.

At each projected gradient descent step, the perturbation update is therefore modulated by this weighting.

Let $\delta_Y^{(t)}$ denote the current perturbation (DCT domain) and $L(\delta_Y)$ the adversarial loss. The update will use a step size η and momentum coefficient β . The per-iteration update will be:

$$\begin{aligned} g^{(t)} &= \text{sign}(\nabla_{\delta_Y} L(\delta_Y^{(t)})) \odot W_{freq} \\ m^{(t)} &= \beta m^{(t-1)} + (1 - \beta) g^{(t)} \\ \delta_Y^{(t+1)} &= \text{Proj}_{\|\cdot\|_\infty \leq \epsilon}(\delta_Y^{(t)} + \eta m^{(t)}) \end{aligned}$$

Here W_{freq} is the continuous frequency-weight, $\odot$ denotes element-wise multiplication, and $\text{Proj}_{\|\cdot\|_\infty \leq \epsilon}(\cdot)$ enforces a global perturbation magnitude constraint. This formulation explicitly reallocates the effective perturbation budget across DCT frequency bands by scaling gradient updates during optimization, thereby biasing the perturbation toward frequency regions that depending on the selected W_{freq} .

Putting these elements together, the constrained optimization problem can be written as:

$$\delta_Y^* = \arg \min_{\delta_Y} \|E(\text{JPEG_D}(\alpha_Y + \delta_Y, \alpha_{Cb}, \alpha_{Cr}; Q_{alg}))\|_2^2$$

$$\text{subject to } \|\delta_Y\|_\infty \leq \epsilon,$$

$$\text{updates applied as } \delta_Y \leftarrow \delta_Y + \eta \cdot m$$

The perturbation is added **after the quantization step** in the JPEG pipeline. This allows gradients $\nabla_{\delta_Y} L$ to backpropagate through *JPEG_D* without passing through the non-differentiable quantization function, which would otherwise zero out gradients.

Directly solving this problem at full (512×512) resolution remains challenging due to the high dimensionality of δ_Y and the non-convex nature of the latent-space objective. To address this high-dimensional optimization difficulty, the proposed algorithm organizes the optimization into a multi-scale hierarchy. For each scale $s \in \{0.25, 0.5, 1.0\}$, we define scaled versions of the original and protected images:

$$x_s = \text{RESIZE}(x, s), \quad x'_s(\delta_Y) = \text{RESIZE}(x'(\delta_Y), s),$$

and consider the corresponding scale-specific latent loss

$$L_s(\delta_Y) = \|E(x'_s(\delta_Y))\|_2^2.$$

Optimization proceeds from coarse to fine scales, warm-starting each finer scale from the perturbation obtained at the previous, coarser scale. This hierarchical scheme reduces the effective search space at early stages, improves initialization quality at full resolution, and provides a natural mechanism for stabilizing optimization.

Finally, in order to handle the local minima problem, the constrained optimization is done by the projected gradient descent method applied with momentum and periodic cold resets (the momentum and the step size are reset according to a predetermined schedule, while the current perturbation is constant or “frozen”). This not only keeps the best solution found but also refreshes the optimizer’s state which allows getting off the plates without going against the magnitude, quantization, or frequency-weighted update logic. Thus, the warm starts upscaling and full-resolution cold resets together are to be termed as the “warm-start, cold-reset algorithm”, the proposed optimization strategy. Consequently, the operational flow of this framework shown in Figure 6 illustrates how these components are integrated during the implementation.

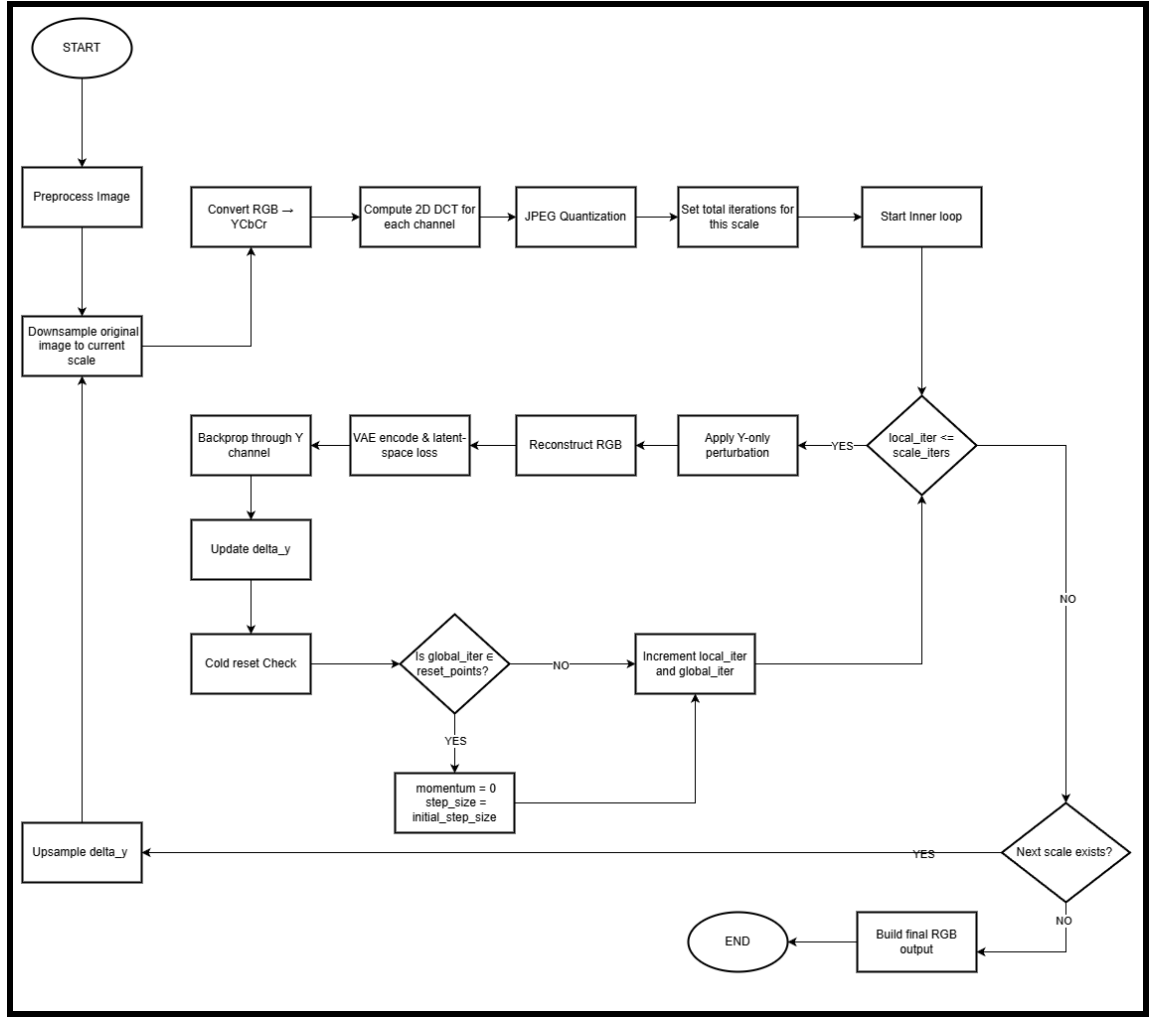


Figure 6

Flowchart of the DCTroy Perturbation Process.

Core Design Principles

This section will highlight the main design principles that will be applied in creating the suggested approach. The principles are based on knowledge gained from studies of frequency-domain defenses, perceptual signal processing, and

optimization. Each principle signifies a conscious design decision that the researchers will make while developing the proposed method, taking into account the needs of the editing threat model used in this research.

Applying Perturbation in DCT Domain

The first principle of the design will be the application of perturbations in the JPEG discrete cosine transform (DCT) domain directly, rather than in the pixel space. The proposed method, by applying perturbations to quantized 8×8 DCT coefficients, will take into account explicitly the way perturbations are disseminated through the JPEG processing pipeline. DCT domain perturbation allows the manipulation of each frequency band and color channel in a very specific way, which is different from pixel-space methods. By doing this, the result will be more efficient, and noise imperceptibility can be minimized.

Color-Channel Prioritization

The focus of the second design principle is on applying the perturbation to a specific color channel to reduce computational cost. The purpose of this design principle is to prioritize the color channel that carries most of the image's structural and semantic representation, which is usually the luminance (Y) channel in the YCbCr color space. Furthermore, this channel also tends to be more resilient to lossy operations, such as JPEG compression. At the same time, this design principle avoids applying perturbation in the chrominance channels, so that the image's color appearance remains unchanged, since it might be removed during chroma downsampling anyway.

Frequency Band Weighting

The third design principle is about taking into account that the different frequency bands in a DCT block representation of the image - the low-, mid-, and high-frequency coefficients - can be affected differently. Hence, the design principle is about selectively applying the perturbation using a continuous frequency band weighting to balance the impacts on edit-protection strength, visual imperceptibility, and overall robustness by allocating perturbation energy according to the specific frequency band, thus creating a structured and perceptually consistent distribution of perturbations for the proposed approach.

Warm-Start, Cold Reset

The fourth design principle introduces the warm start, cold reset approach. To address the challenges of initialization sensitivity and local minima in complex loss landscapes, the framework incorporates warm-start upscaling and periodic cold-resets. The warm-start strategy initializes higher-resolution optimization stages using upsampled versions of previously perturbed images, providing a meaningful starting point that prevents the optimizer from beginning in uninformative regions and ensures better convergence at the target resolution. Complementing this, periodic cold-resets improve exploration by "freezing" the current perturbation while re-initializing optimizer states, such as momentum buffers and step sizes. This mechanism removes accumulated inertia and historical bias that can lead to stagnation in flat regions, effectively restarting the search with fresh parameters while preserving all prior progress.

Differentiable JPEG Pipeline

The fifth design principle will focus on utilizing the differentiable JPEG Pipeline to ensure the generated perturbations are robust against compression-based purification methods. Standard JPEG compression involves non-differentiable operations, particularly quantization and rounding, which impede the backpropagation of gradients and prevent the optimizer from accounting for compression artifacts. By integrating a differentiable approximation of the JPEG encoding and decoding process into the optimization loop, the framework allows gradients to flow through the compression steps. This mechanism enables the generation of "quantization-aware" perturbations that can survive the frequency masking and precision loss inherent to JPEG compression, thereby maintaining their effectiveness even when the image is subjected to purification techniques.

Evaluation

In accordance with the Design Science Research Methodology, the researchers utilized the FEDS framework as the basis for the overall evaluation strategy. Therefore, the research chose to apply a formative evaluation strategy involving artificial means. Specifically, a major aspect of this approach is the continuous improvement of the artifact in a controlled setup to ensure technical effectiveness before introducing it to real-world conditions.

Experimental Setup

In order to critically evaluate the suggested DCT-based adversarial perturbation framework, the researchers have set up a configuration of targeted experiments, each explicitly corresponding to a specific goal: edit protection effectiveness, noise imperceptibility, robustness, and convergence efficiency. Thus, for each experiment, the researchers clearly indicate the target objective, list the datasets and editing models involved, mention the baseline comparison methods and hardware used, and enumerate all evaluation metrics and criteria. Accordingly, this methodical process guarantees that the results presented correspond exactly to the previously defined success factors, making it possible to conduct a transparent and quantitative comparison with other techniques.

Benchmark Datasets

For the purpose of making sure that edit protection gets evaluated under varied and difficult real-world scenarios, this research employs two of the most recognized benchmark datasets, namely: InpaintCOCO and OmniEdit.

- **OmniEdit (Dataset A):** is known to be a benchmark dataset for semantic facial editing, which provides very good quality pictures of human faces together with the edits comprising changes in attributes, alterations in expressions, and identity-related transformations, as well as the appropriate text prompts; its uniform structure allows for the comparison of img2img generative models throughout the evaluation process.

- **InpaintCOCO (Dataset B):** a wide-ranging dataset offering segmentation masks, bounding boxes, random polygons, and random boxes, as well as text-guided and unguided prompts, which makes it a perfect choice for controlled inpainting-based editing tasks.

For this study, a sampling of 500 images from each dataset (1000 altogether) is selected to be used in the controlled img2img and inpainting tests. Furthermore, the chosen images are all resized to 512×512 to unify input resolution and guarantee the evaluation done is consistent and comparable under all experimental conditions.

Baselines

In order to put the performance of the proposed method into context, the researchers will conduct a comparison with well-known adversarial protection methods whose main objective is the same, namely, introducing perturbations to images so as to prevent unauthorized or malicious generative edits. Specifically, the following baseline methods are mentioned: PhotoGuard, Mist, and DiffusionGuard. Particularly, the reason behind the selection of these methods is that they all constitute typical applications of perturbation-based defense in the context of diffusion-based image editing, and their effectiveness has been empirically demonstrated.

Threat Model

The researchers speculate that a malicious actor could use an off-the-shelf latent diffusion model (LDM) which was pre-trained beforehand, with the Variational Autoencoder (VAE) component considered as a frozen module, to carry out the unauthorized or malicious edits to the protected image. The researchers prepare this protected image ready by applying the adversarial perturbations on the original picture through the proposed method. The researchers have knowledge of neither the exact U-Net architecture nor the model weights that the malicious actor has, nor the editing prompt that has been used to carry out the intended edit. This makes a very realistic gray-box threat scenario.

- **InstructPix2Pix**: a model that is based on the Stable Diffusion architecture and can do many different semantic edits according to the input of natural language like changes of facial features, alterations of emotions, and even to the level of transformations of objects. The model's general popularity and versatility have made it a good contender for carrying out surefire instruction-based editing attacks on identity-relevant issues
- **Stable Diffusion Inpainting**: a tool that focuses on the high-quality removal, replacement, and restoration of regions with the help of binary masks. This feature allows for the testing of mask-guided inpainting attacks that pose a significant threat to security in practice.

This guarantees that the two main editing paradigms—**instruction-based semantic editing and mask-based inpainting**—will be fully covered, allows

direct comparison with the existing baselines, and establishes a strict evaluation of the cross-model robustness and generalizability of the framework proposed.

Hardware Environment

All of the experiments will be executed in the same runtime environment, specifically:

- GPU: NVIDIA GeForce RTX 4090, 24GB VRAM
- CPU: Intel(R) Core(TM) i9-14900KF
- Memory: 32 GB system RAM

Software Environment

The implementation will be developed mainly using Python, and it made use of various libraries, including:

- **PyTorch**: Deep learning framework for VAE/diffusion models
- **NumPy/SciPy**: Numerical computation for DCT operations and statistical analysis
- **PIL/Pillow**: Low-level image I/O and JPEG quality control
- **scikit-learn**: Statistical analysis, metric computation
- **scikit-image (skimage)** – Computing low-level image quality metrics.
- **Matplotlib/Seaborn**: Data visualization, result plots
- **lpips** – Implemented the LPIPS metric using pre-trained AlexNet features.
- **clip-score**: Used to compute CLIP-based semantic consistency scores

Test Environment

The test environment will consist of:

- A controlled workflow for experiment orchestration
- Fixed seeds for random number generators where applicable
- Consistent preprocessing pipelines for all input images

This uniform test environment ensures that all experimental results stem from a consistent and fully reproducible setup.

Evaluation Plan

The evaluation plan shown in Figure 7 consists of three phases structured as an ablation-style evaluation to isolate the contribution of each component of the proposed framework. Initially, **Phase 1** is divided into two parallel analyses: (a) Frequency Band Weighting, which examines how perturbation distribution across low-, mid-, and high-frequency bands affects edit protection, noise imperceptibility, and robustness using a fixed PGD optimizer; and (b) Optimization Strategy Evaluation, which assesses the impact of the warm-start and cold-reset algorithm as the optimizer without frequency weighting. Thereafter, **Phase 2** integrates these components into the full proposed method and evaluates it against the two Phase 1 variants to actually evaluate how each module contributes to overall performance. Finally, **Phase 3** conducts a comparative

benchmark between the proposed method and state-of-the-art baseline defenses to determine its relative effectiveness and practical robustness. For clarification, Dataset A is used for img2img while Dataset B is used for inpainting. It is also important to note that for the inpainting type of diffusion-based image editing, CLIPScore will replace LPIPS as the edit protection effectiveness metric.

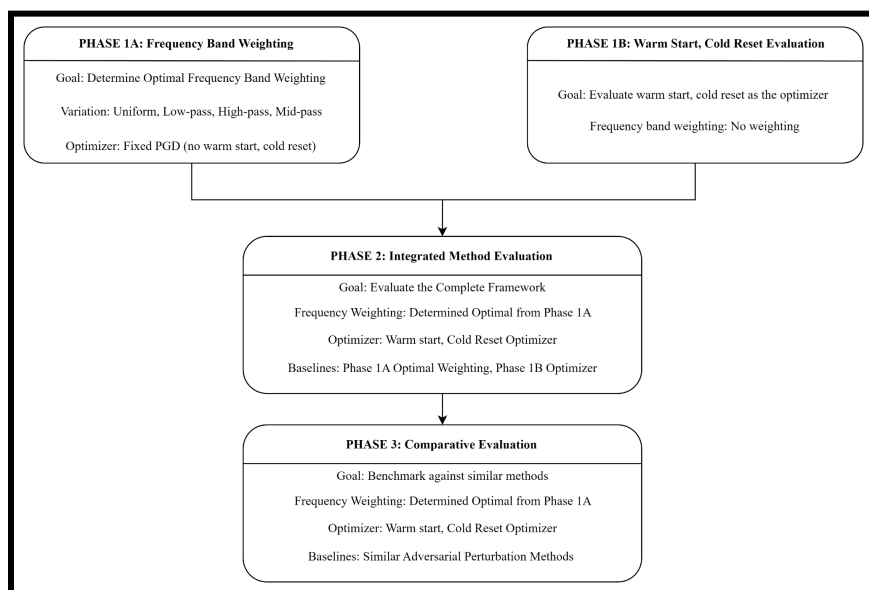


Figure 7

Multi-Phase Evaluation Plan

Phase 1A: Frequency Band Weighting

In this part of the first phase, the study will explore how the distribution of perturbations across frequency bands affects edit protection effectiveness, noise imperceptibility, and robustness. To begin with, a fixed PGD optimizer will be used without warm-start or cold-reset strategies. Then, frequency weights will be

systematically varied across DCT bands using four representative patterns: uniform (no decay), low-pass (decay towards high frequencies), high-pass (decay towards low frequencies), and mid-pass (decay around mid frequencies). Consequently, the experiments will use 20% of Dataset A and 20% of Dataset B. Thereafter, the evaluation metrics will include SSIM, PSNR, and VIF for noise perceptibility, LPIPS for edit protection quality, and LPIPS degradation after common distortions to measure robustness. Finally, the expected outcome is to identify frequency weighting configurations that optimally balance imperceptibility, robustness, and protection quality.

Phase 1B: Optimization Strategy Evaluation

Following this, the second part of the first phase will investigate the combined effect of warm-start and cold-reset optimization strategies on perturbation quality and convergence efficiency. Firstly, standard frequency weights (no weighting) will be applied, and the dataset will include 20% of Dataset A and 20% of Dataset B, distinct from Phase 1A. Then, the study will assess how the combined optimizer impacts convergence by monitoring relative convergence rate to evaluate computational efficiency. Ultimately, this phase will quantify the contribution of the optimization strategy to perturbation quality and efficiency.

Phase 2: Integrated Method Evaluation

The second phase will evaluate the complete perturbation framework, combining the optimal frequency weighting identified in Phase 1A with the optimization strategy from Phase 1B. As a starting point, Experiments will be conducted on 30% of Dataset A and 30% of Dataset B, which were not used in previous phases. Building on this, the baselines will include the Phase 1A configuration (best frequency weighting with standard optimizer) and the Phase 1B configuration (uniform frequency weighting with warm-start and cold-reset optimizer). Consequently, the evaluation will again use SSIM, PSNR, and VIF for noise perceptibility, LPIPS for edit protection quality, LPIPS degradation for robustness, relative convergence rate for measuring convergence efficiency. Therefore, this phase is expected to demonstrate the effectiveness of the integrated method in achieving high protection quality, imperceptibility, and computational efficiency relative to the two isolated components examined in Phase 1.

Phase 3: Comparative Evaluation

Finally, the final phase will benchmark the proposed integrated method against comparable similar perturbation methods. To begin with, using the optimal configuration from Phase 2, experiments will employ 30% of Dataset A and 30% of Dataset B. Thereafter, the baselines will include existing frequency-domain

perturbation methods and other inference-time protection approaches. Subsequently, the evaluation metrics will remain consistent, assessing noise perceptibility (SSIM, PSNR, VIF), edit protection quality (LPIPS), robustness (LPIPS degradation under practical transformations), and convergence efficiency (Relative Convergence Rate). Ultimately, this phase aims to establish the relative performance of the proposed method and its practical applicability in defending against instruction-based semantic editing and mask-guided inpainting attacks.

Experimental Procedures

The researchers put together the design for the experiments that will be conducted in the study, one for each goal, to evaluatively and systematically examine if the proposed method satisfies its preset success criteria by using controlled procedures for showing the effectiveness quantitatively and obtaining information for iterative refinement.

Edit Protection Effectiveness Evaluation

To begin with, the researchers will assess how well the artifact was able to prevent semantic similarity between edits on both the protected and original images, which was in line with the Edit Protection Effectiveness Objective.

The method that will be used in the experiment, which is depicted in Table 3, will use a controlled setup that separates the influence of the perturbations, thus, ensuring that the difference between edited outputs reflects the effect of the

proposed method. Then, the perceptual divergence will be measured by LPIPS or CLIPScore that indicates the semantic dissimilarities much better than the pixel-space metrics, hence, the visual changes are represented. Thereafter, the results will be consolidated over all the test samples and against the already-set baselines, thus, a clear indication of the protection capacity is provided. Ultimately, the researchers address the two primary editing paradigms by incorporating both instruction-based semantic editing (InstructPix2Pix) and mask-based inpainting (Stable Diffusion Inpainting 1.0), enabling evaluation of cross-model transferability, and comparative effectiveness relative to existing defenses.

Table 3

Edit Protection Quality Evaluation Procedure

	Action	Input(s)	Output(s)
1	Load Images	- Original Image - Protected Image	Loaded (512×512)
2	Edit Original	Original + Edit Prompt	Edited Original
3	Edit Protected	Protected + Same Edit Prompt	Edited Protected
4	Compute LPIPS or CLIPScore	- Edited Original - Edited Protected	LPIPS or CLIPScore Score
5	Aggregate Results	All LPIPS scores	Mean, Standard Deviation

Noise Imperceptibility Evaluation

Building on this, this experiment aims to determine if the suggested artifact keeps the visual quality of protected images with the introduction of perturbations within the images, which was aligned with the Perturbation Imperceptibility Objective suggests.

The controlled experimental procedure outlined in Table 4 guarantees to get exact quantitative measurements of the changes made by the artifact while keeping all the test samples consistent. Consequently, to evaluate the imperceptibility of the perturbations, the researchers will measure SSIM (structural similarity), PSNR (pixel-space fidelity), and VIFp (visual information fidelity) for each protected image and its corresponding original one, thus enabling them to give a collective estimate of the extent to which the protected images still contain the structural content, perceptual quality, and visual information of the original. Finally, the results will be combined for all test samples, and for each metric, the mean and standard deviation are computed.

Table 4

Perturbation Imperceptibility Evaluation Procedure

	Action	Input(s)	Output(s)
1	Load Images	- Original Image	Loaded (512×512)
2	Apply Protection	Original Image + Protection Artifact	Protected Image
3	Compute SSIM, PSNR, VIFp	- Original Image - Protected Image	SSIM, PSNR, and VIFp scores

Table 4*Perturbation Imperceptibility Evaluation Procedure (cont'd)*

	Action	Input(s)	Output(s)
4	Aggregate Results	All SSIM, PSNR, and VIFp scores	Mean, Standard Deviation

Robustness Quality Evaluation

On the other hand, the goal of this particular experiment is to check if the suggested method continues to be effective against attacks during image transformations and purifications which correspond to the Robustness Objective. Hence, the experimental process, which is briefly shown in Tables 5 and Table 6, separates two different situations: JPEG compression, and AdvClean purification. Each scenario represents a typical image processing or image quality purification through which the artifact would still be able to protect against editing and thus validating if the proposed method preserves edit protection despite typical real-world or adversarial modifications.

Table 5 evaluates the robustness of the protection artifact against **JPEG compression**.

Table 5*JPEG Compression Robustness Evaluation Procedure*

	Action	Input(s)	Output(s)
1	Load Images	- Original Image - Protected Image	Loaded (512×512)
2	Compute LPIPS or CLIPScore	- Edited Original - Edited Protected	LPIPS or CLIPScore Score
3	Apply JPEG Compression	Protected Image + JPEG Compression	JPEG Compressed Protected Image
4	Compute LPIPS or CLIPScore degradation	- Edited JPEG Compressed Protected - Edited Protected - Edited Original	LPIPS or CLIPScore Degradation Score
5	Aggregate Results	All LPIPS or CLIPScore Degradation scores	Mean, Standard Deviation

Initially, the baseline LPIPS or CLIPScore will be computed to determine the protection efficiency before the original images are loaded and edited. Afterwards, protected images are compressed at varying quality rates, and the post-compression LPIPS or CLIPScore is assessed in relation to both the edited originals and the edited protected images. Finally, degradation scores are summed over all images, thus allowing the researchers to determine how effectively the artifact maintains edit protection during lossy compression.

The procedure in Table 6 evaluates the robustness of the proposed method under **AdvClean purification**.

Table 6

Purification Robustness Evaluation Procedure

	Action	Input(s)	Output(s)
1	Load Images	- Original Image - Protected Image	Loaded (512×512)
2	Compute LPIPS or CLIPScore	- Edited Original - Edited Protected	LPIPS or CLIPScore Score
3	Apply AdvClean Purification	Protected Image + Purification	Purified Protected Image
4	Compute LPIPS or CLIPScore degradation	- Edited Purified Protected Image - Edited Protected - Edited Original	LPIPS or CLIPScore Degradation Score
5	Aggregate Results	All LPIPS or CLIPScore Degradation scores	Mean, Standard Deviation

To begin with, the original and protected images are first loaded and edited to get baseline LPIPS or CLIPScore, whereby the effectiveness of protection is established before purification. Thereafter, the protected images are then purified with AdvClean, and the post-purification LPIPS or CLIPScore is measured relative to both the edited originals and the edited protected images. Lastly, the degradation scores are then collected for all the images, and the mean and standard deviation are computed. Ultimately, this design provides a way to quantitatively

evaluate how well the artifact can retain its edit protection when faced with purification countermeasures.

Convergence Efficiency Evaluation

Additionally, it is the aim of this study to determine if the protection artifact shows better convergence in the quality of edit protection by using the proposed warm-start and cold-reset algorithm in line with the Convergence Performance Objective. The experimental procedure, which is summarized in Table 8, involves taking LPIPS or CLIPScore scores at the intermediate iterations (50, 100, 500, 1000) to measure how fast the artifact gets to the target semantic divergence threshold ($\text{EDIT PROTECTION METRIC} \geq 0.63$). Subsequently, computation time per iteration is recorded to give a better understanding of the algorithm's efficiency which will then be used for the calculation of the relative convergence rate.

Table 8

Convergence Performance Evaluation Procedure

	Action	Input(s)	Output(s)
1	Load Images	- Original Image - Protected Image	Loaded (512×512)
2	Apply Protection	- Original Image + Protection Artifact	Protected Image

Table 8*Convergence Performance Evaluation Procedure (cont'd)*

	Action	Input(s)	Output(s)
3	Compute Relative Convergence Rate	- Edited Original - Edited Protected at iterations 50, 100, 500, 1000	LPIPS or CLIPScore Scores per Iteration
4	Aggregate Results	All LPIPS or CLIPScore scores at each iteration	Mean, Standard Deviation per iteration
5	Compute Computation Time	Time per iteration	Iteration Time / Total Time

The method outlined in Table 8 assesses the efficiency of convergence of the suggested artifact. First, the original and protected images are prepared, and LPIPS or CLIPScore is calculated at the defined intermediate iterations to measure the semantic divergence between the edited originals and the edited protected images. Afterwards, for each iteration, the computation time is noted to give a performance context in real-time. Then, the LPIPS or CLIPScore scores are summed up for all the test samples, and the mean along with the standard deviation is calculated for each iteration. Thereafter, using these measured values, the relative convergence rate will be calculated. Thus, the experimental setup allows for a rigorous and quantifiable evaluation of the convergence nature of the proposed warm-start, cold-reset algorithm.

Evaluation Metrics

A set of evaluation metrics has been utilized to assess to what extent the proposed method has achieved its goals. These metrics are grouped into different categories: **edit protection effectiveness, noise imperceptibility, robustness, and convergence efficiency**. Each of these metrics is designed to measure one particular aspect of the performance of the artifact and corresponds to the success criteria for the objective in question. It was also considered that these metrics would allow for a just comparison with the baseline methods.

LPIPS (Learned Perceptual Image Patch Similarity)

- Quantifies perceptual distance between two images by comparing deep convolutional features.
- Captures semantic and perceptual differences beyond pixel-level variations, ideal for assessing divergence between edits on protected versus original images.
- A high LPIPS score indicates substantial visual divergence, demonstrating effective disruption of identity-preserving edits.
- Computed using the **lpips Python library with pre-trained AlexNet** features.

SSIM (Structural Similarity Index)

- Quantifies structural similarity between original and protected images by comparing luminance, contrast, and structure.

- Values range from -1 to 1; closer to 1 indicates higher similarity.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Where μ_x, μ_y means, σ_x^2, σ_y^2 are variances, σ_{xy} is the covariance, and C_1, C_2 are small constants.

- Ensures that visual content remains recognizable after perturbation.
- Computed using **skimage.metrics.structural_similarity** with `channel_axis=2` for RGB images.

PSNR (Peak Signal-to-Noise Ratio)

- Measures pixel-level fidelity by comparing the reconstructed image to the reference in logarithmic decibel space.

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

Where MAX_I is the maximum possible pixel value, and MSE is the mean squared error between the reconstructed image I_{SR} and the ground truth I_{HR} :

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N (I_{HR}(i,j) - I_{SR}(i,j))^2$$

- Higher PSNR values indicate lower pixel-space distortion, making the metric suitable for assessing the imperceptibility of perturbations.
- Computed using **skimage.metrics.peak_signal_noise_ratio** with images scaled to the [0,255] range.

VIFp (Visual Information Fidelity, pixel-domain)

- Quantifies how much visual information from a reference image is preserved in a distorted version
- It evaluates information fidelity making it sensitive to structural and perceptual distortions that simple pixel-wise metrics cannot capture.

$$VIFp(x, y) = \frac{\log_2 \prod_b (1 + g_b SNR_b^{distorted})}{\log_2 \prod_b (1 + SNR_b^{reference})}$$

Where the **numerator** represents the mutual information between the distorted image and the observer's internal representation and the **denominator** represents the mutual information for the original (reference) image.

- Higher VIFp scores indicate stronger preservation of perceptual information, while lower scores signify that the transformation or attack has removed or altered significant visual detail.
- Computed using the **piq Python library (PyTorch Image Quality)**, which implements the full-reference VIFp metric across four wavelet scales. Inputs are normalized Torch tensors.

CLIP Score (Contrastive Language-Image Pre-training)

- Measures edit protection quality by quantifying the semantic separation between the original and protected images.

- Computes the cosine similarity between the feature vector of the original image (f_o) and that of the protected image ($f_{\square}$) embedded in the CLIP multimodal space (e.g., ViT-B/32).
- To reflect semantic divergence, the score is calculated as $1 - \text{cosine similarity}$. A higher value indicates stronger protection against identity-aware edits.
- Typically computed using the **clip-score** library with a pre-trained CLIP model.

LPIPS Degradation (Learned Perceptual Image Patch Similarity)

- Evaluates robustness of the perturbation under controlled transformations.
- Calculates the perceptual distance (LPIPS) before and after the transformation, then measures the percentage drop:

$$\text{Degradation} = ((\text{LPIPS_before} - \text{LPIPS_after}) / \text{LPIPS_before}) \times 100\%$$

LPIPS itself is computed as a weighted Euclidean distance between deep network feature activations (commonly using the AlexNet backbone).

- Lower degradation indicates better robustness and stronger persistence of the adversarial effect.

Relative Convergence Rate

- Quantifies computational efficiency by measuring how quickly the optimization achieves near-optimal protection.
- Tracks the number of iterations (t) required to reach a predefined fraction (e.g., 90%) of the final protection score (L_{final}):

$$\text{Steps at 90\%} = \min (t \mid L(t) \geq 0.90 \times L_{\text{final}})$$

Here, $L(t)$ can correspond to LPIPS, CLIP divergence, or a composite metric used for protection evaluation.

- Fewer steps indicate faster convergence and more efficient optimization.

Runtime Overhead

- Measures computational cost as the **average execution time per image**.
- Let T_i denote the total execution time required to process image i , including all defense-related operations. Runtime overhead is defined as:

$$\text{Runtime Overhead} = \frac{1}{N} \sum_{i=1}^N T_i$$

where N is the number of evaluated images.

- Lower values indicate more computationally efficient defenses and greater suitability for large-scale or real-time deployment.

GPU Memory Overhead

- Measures memory cost as the **peak GPU memory usage during execution**.
- Let M_i^{peak} denote the maximum GPU memory allocated at any point while processing image i . GPU memory overhead is defined as:

$$GPU\ Memory\ Overhead = \max_{i \in \{1, \dots, N\}} M_i^{peak}$$

- Peak memory usage is recorded using the framework’s memory profiler (e.g., PyTorch CUDA memory statistics).
- Lower peak memory usage indicates better scalability and compatibility with memory-constrained hardware.

Ethical Considerations

The ethical issues surrounding the employment of human-related image data and adversarial perturbation techniques were very crucial in this research and they need at all times the most careful thought. The subsequent points describe the rules and practices that were followed in order to maintain the legal and moral principles during the creation and assessment of the suggested framework.

Dataset Usage

The datasets that have been used in this research, namely OmniEdit and InpaintCOCO, are open-access public research datasets. These datasets are available for academic use and redistribution, as well as modification, according to the terms of their respective licenses. The researchers made sure to check the licensing terms so that they could use the data properly and ethically.

Privacy and Human Subjects

Researchers had exclusive access to all data which were safely stored in the experiment environment. The images were for the sole purpose of assessing the proposed framework and no commercial, personal or non-research purposes were considered. The dual-use of adversarial perturbations is, however, recognized by the researchers. To use materials responsibly, the research is entirely defensive-application-oriented, which means it is aimed at protecting people from unauthorized or harmful image edits. There is no development or support of any offensive or misuse-enabling techniques.

All in all, the study follows ethical research standards that are in place, making certain that there is privacy, use of the dataset is responsible, and human-related image data is properly handled.

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